00:00
The time to relax, chill.
是时候放松、冷静了。

00:03
In break, maybe, and all the leaders are struggling right now.
课间休息吧，所有的领导现在都在挣扎。

00:10
This week is kind of a watch point. Okay, so this is what we're going to do today. We will, as usual, run through a couple of blocks from last session. The last session we did was our group presentations, right?
这周是一个观察点。好了，这就是我们今天要做的。像往常一样，我们将回顾上节课的几个街区。我们上节课做的是小组报告，对吧?

00:21
You're working progress presentations. And I think reading what everybody kind of took away from the session, it was very clear that you're all learning from each other.
你在做进度报告。我认为阅读每个人从会议中得到的东西，很明显你们都在相互学习。

00:32
In terms of seeing what kind.
在看什么样的方面。

00:33
Of data your friends are working on, what kind of topics your friends are working on, what kind of projects they're using, right? There were a lot of common issues that popped up, mostly in terms of the scope.
你的朋友正在研究的数据，你的朋友正在研究什么样的主题，他们正在使用什么样的项目，对吧？出现了许多常见的问题，主要是在范围方面。

00:44
And thinking through, sort of the use of popular tools or popular methods,

right? So that's always, always find that this week, six presentations tend to be.
思考一下，使用流行的工具或者流行的方法，对吧？所以这周的六场演讲。

00:56
Really help sharpen or give you a direction and the way you're going to go. So I'm going through the peer reviews at the moment, and they should.
真的能帮助你明确方向，给你指明前进的方向。所以我现在正在看同行评议，他们应该。

01:03
Be back with you.
再见。

01:06
Has the.
有。

01:07
A visual nonsense. I think so.
视觉上的无稽之谈。我想是的。

01:12
Okay.
好吧。

01:13
Well.
好。

01:16
A little bit of of a and then it's a black box. In the beautiful black box, we'll talk a little bit about what's going into. Jenny.
一点点，然后它是一个黑盒子。在这个漂亮的黑盒子里，我们会讨论一下将要发生的事情。珍妮。

01:26
We should meet you today, but you will continue. But that's okay. I'll go talk a little bit about how it's designed, what it means, or what.
我们应该今天见你，但你得继续。不过没关系。我会讲一下它是如何设计的，它是什么意思。

01:35
We can do as historians with a tool like that, right? Because it's a very different kind of tool in some way to what we have been looking at for the semester so far.
作为历史学家，我们可以用这样的工具，对吧？因为在某种程度上，它是一种非常不同的工具与我们这学期以来一直在研究的工具不同。

01:45
So we'll talk about, yes, it's a quering and a description.
所以我们会讨论，是的，这是一个疑问和描述。

01:51
And do a little exploration.
做一些探索。

01:53
To see how some things work. And then we take a break, and then we talk about recognition of objects.
看看有些东西是怎么工作的。然后我们休息一下，然后我们讨论物体的识别。

02:00
With.
与。

02:02
Current publicly available.
目前可公开获得。

02:06
When you do an exercise for that book, you will talk a little bit about what that looks like, what it means.
当你们为那本书做练习的时候，你们会讨论一下它是什么样子的，它是什么意思。

02:13
And how we can solve that. And then we'll end up with a discussion about quantitative history, which I know is a bit of an intimidating topic to many of us.
以及如何解它。最后我们会讨论定量历史，我知道这个话题对很多人来说有点吓人。

02:24
We don't usually do anything, but it's useful, you know, and actually allow parallels to.
我们通常不做任何事情，但它很有用，你知道，实际上允许并行。

02:33
All right. So one of the questions that seems quite recurrent, I guess, is this concern, or maybe even anxiety, about?
好吧。所以我想，其中一个经常出现的问题是，这种担忧，甚至是焦虑，是关于什么？

02:45
How do we move from a data set to an analysis or an argument, right? And that's very common. All right. I mean, the field of the whole construct was a little bit this question.
我们如何从数据集过渡到分析或论证，对吧？这很常见。好吧。我的意思是，整个建构的领域有点像这个问题。

02:54
Right? What's the difference between sort of accumulating data, structuring data and working tools and the gulf between facts and sort of a

historical argument?
对吧?积累数据，构建数据和工作工具之间的区别是什么？事实和历史论证之间的鸿沟是什么?

03:05
What we think about is you need to refocus yourself on your research question that you were interested in in the first place, because you don't come across the data.
我们认为你需要把注意力重新集中在你最初感兴趣的研究问题上，因为你没有遇到数据。

03:15
Willingly, usually, right? Usually you come across video that because it has some sort of impact in terms of your question or your interests. So then it comes back to you as a historian, right? What I mean?
通常是心甘情愿的，对吧？通常你会遇到这样的视频，因为它对你的问题或兴趣有某种影响。那么作为一个历史学家，这又回到了你身上，对吧？我是什么意思?

03:28
Interested in looking at, if you're interested in seeing it in the dimension of tree, E.G.?
如果你对树的维度感兴趣，比如说?

03:34
Right? And you have a catalogue in your clear set of treat, volumes and goods, maybe, right? Does this shed like on the questions that you're interested in?
对吧?你有一个目录在你的一套明确的待遇，数量和商品，也许，对吗？这能回答你感兴趣的问题吗?

03:42
Conversation, right? Is it properly described or structured in such a way that you can actually ask good questions up of it, right? That will be put in as we go.

谈话,对吗?它的描述和结构是否恰当，能让你提出好的问题，对吧？我们会把它放进去。

03:53
So.
所以。

03:53
I'll be describing the fields of particular way. Is there a way to kind of add in or to?
我会用一种特殊的方式来描述这些场。有没有一种方法可以加进去或者加到?

04:00
Use the structure it has right now to as a proxy for gender, E.G., are they public goods that are more gender? Are they publicly trades and more gender public sectors?
用它现在的结构作为性别的代表，例如，它们是更性别化的公共产品吗？它们是公共贸易还是性别平等的公共部门?

04:10
Fun, little more gender and using and sort of imagination, how to bridge between the data set and the question itself.
有趣的是，更多的性别，使用和想象，如何在数据集和问题本身之间架起桥梁。

04:20
Right? It's that is starting to look. It's not something that amission can do for you, right?
对吧?这是开始看。这不是任务可以为你做的，对吧?

04:26
And this is where you come in, as I saw it.
在我看来，这就是你的切入点。

04:29
Sometimes walking from the other way around. You can look at the data set, and sometimes questions just jump up at you, right? So you look at E.G., say.
有时反过来走。你可以看看数据集，有时问题会突然出现，对吧？比如说，你看一下e.g.。

04:42
What they just said about property prices and mart stations in Singapore, right? So how close, what is the pattern between sort of, how expensive a property is and how close it is to.
他们刚才说的新加坡的房价和商场，对吧？那么，一个房产的价格和它的价格之间的关系是怎样的呢。

04:55
Multi station, right? Constant movie in Singapore, right? So that from that data set, you can look at it, look at what kind of, sort of attributes it has, what kind of additional information it has.
多站，对吗？在新加坡经常看电影，对吧？从数据集中，你可以看到它，它有什么样的属性，它有什么样的附加信息。

05:07
And maybe you can also see, hey, what about distance to schools? Right? If it's recorded, there can use an analysis and compare between different factors.
也许你还可以看到，嘿，到学校的距离呢？对吧?如果有记录，就可以对不同因素进行分析和比较。

05:17
Sorry, how about you isolate one specific factor, sort of influence on the prices?
不好意思，你能不能单独考虑一个影响价格的因素?

05:23
Um, or if I don't know if you want to bring in.

或者我不知道你要不要带。

05:27
If you have a database of, say, who's buying these flags, right? Recent controversy, who is buying these flats, and where do they work?
如果你有一个数据库，比如说，谁在买这些旗帜，对吧？最近的争议是，谁在买这些公寓，他们在哪里工作？

05:40
Right? That's the common analysis that comes jumps, that you are the questions that come up at you, right? As you, as a historian, are investigating a data, so you may not have gone into it with.
对吧?这是跳出来的常见的分析，你就是出现在你面前的问题，对吧？作为一名历史学家，你在调查数据，所以你可能没有深入研究。

05:51
Good question, but it seems like this data set can support that kind of argument or the kind of investigation then you.
好问题，但似乎这个数据集可以支持那种论点或那种调查，然后你。

05:58
The suit. But that's generally the two reason. If you do research, do you do the same thing with sources like traditional sources? You go into an archive, you take off a box of material, you're kind of flipping through it. It's kind of roughly related to your interests. And then it kind of tells you a story, or it suggests.
该诉讼。但这通常是两个原因。如果你做研究，你会对传统资源做同样的事情吗？你走进一间档案室，拿出一盒材料，翻看一遍。这和你的兴趣大致相关。然后它会告诉你一个故事，或者暗示。

06:13
To you a particular thing that you may want to investigate and that might form your research question. Right? All you may really have gone is with a research question, and what you're doing is a drawing truth.
对你来说，你可能想要调查的一个特定的事情，这可能会形成你的研究问

题。对吧?你所做的可能只是一个研究问题，而你所做的是一个绘画真理。

06:24
Looking for things that could give you a.
找一些能给你。

06:28
I definitely see with the data set.
我完全可以从数据集看出。

06:30
No difference, other than thinking about how you work through it and whether you can reuse a data set in a way that you can reuse a source, right? A source is very easy to.
没有区别，除了考虑如何处理它以及是否可以像重用源一样重用数据集，对吧？一个源是很容易的。

06:41
To kind of figure out and adapt to your argument at the other side. It's much harder because you may need to restructure it. You may need to clean it. You may need to do all sorts of software management.
找出并适应对方的观点。这要困难得多，因为你可能需要重组它。你可能需要清洗一下。您可能需要进行各种软件管理。

06:51
With the asset before you can start using it. That's kind of, we need some critical Okay.
在你开始使用资产之前。这有点，我们需要一些关键的。

07:00
No.
否。

07:00
Oh, we get, I mean, if you need to pass somewhere, or there's something

you want to do that, okay.
哦，我们明白，我是说，如果你需要通过某个地方，或者有什么你想做的事，好吧。

07:07
Okay. The second kind of big concern, of course, I think that everybody faces at some point is hope, right? I realized my project's too big, too ambitious.
好吧。第二个大问题，当然，我认为每个人在某种程度上都面临的是希望，对吧？我意识到我的项目太大了，野心太大了。

07:17
There's too much I need to think about, too many variables that I need to consider, right? I know some projects were considering something quite thick, and there were many, sort of moving parts.
我要考虑的东西太多了，我要考虑的变量太多了，对吧？我知道有些项目正在考虑一些非常复杂的东西，而且有很多移动的部分。

07:29
Right? And soon you'll find that you don't actually really have much time to do all of them, right?
对吧?很快你就会发现你实际上并没有太多的时间去做所有的事情，对吧?

07:36
So you need to identify which one we will focus on. And good. Well.
所以你需要确定我们要关注哪一个。和良好的。好。

07:40
When should a whole partner? And when it comes to scoping and deciding what you want to focus on comes, goes back to the same question about the reason.
什么时候应该有一个完整的伴侣？当涉及到范围和决定你想要关注什么时，又回到了同样的问题，关于原因。

07:51

Right? Sometimes you need a little bit of the data exploration, right? You work through cleaning the data, getting the data together, right? And then you see from there.
对吧?有时候你需要一些数据探索，对吧？你要清理数据，把数据收集起来，对吧？然后你从那里看到。

08:02
Kind of what is most promising, and then from there, that helps you become a missile, right? So if, let's say you're looking at, right, you know, metro history data, right?
最有前途的东西，然后从那里，帮助你变成一枚导弹，对吧？假设你在看地铁历史数据，对吧?

08:11
And then it gives you a lot of information about photography, right?
然后它给你很多关于摄影的信息，对吧?

08:15
Then maybe that's a good topic limitation the data set will support this work. So I will pursue this question because right out of the five questions I would be interested in, this is the one that the data sets most of the for, and therefore of the Student.
也许这是一个很好的主题限制，数据集将支持这项工作。所以我将继续研究这个问题，因为在我感兴趣的五个问题中，这是一个数据集最多的问题，因此也是学生的问题。

08:27
Okay, so the data set can constrain or shift your scope for you, right? Or, um.
好的，数据集可以为你限制或转移你的范围，对吧？或者,嗯。

08:33
I won't go any more. But that's generally, that's one way of thinking about how you want to decide the scope and which part of the.
我再也不去了。但一般来说，这是一种思考方式你想要如何决定范围和哪个

部分。

08:43
Look that you're thinking of if you want to make you focusing. And that's why it is really important sometimes to go to the whole packline, right to come.
看看你在想什么，如果你想让自己集中注意力。这就是为什么有时候去整个包装线很重要，马上就来。

08:52
Work through the data set. Don't just spend all your time finding the different data sets and not putting them together and investigating them.
处理数据集。不要把所有的时间都花在寻找不同的数据集上，而不是把它们放在一起进行调查。

09:00
Right, walk through it, create a preliminary data set, and then see what's there, because you might need to go backwards. Reiterate.
对，遍历它，创建一个初步的数据集，然后看看那里有什么，因为你可能需要倒回去。重申。

09:09
We do something clean it in another way, restructure in a popular way. You want to keep it all the way to the end when you realize that, oh, oh, this is not going to it's not going to work.
我们用另一种方式清理它，用一种流行的方式重组。你想一直坚持到最后，当你意识到，哦，哦，这不会不会成功。

09:20
You want to kind of try and reach the end as early as possible, and then see what tricks, see what does it support like, then you have time to go back.
你想要尽可能早地到达终点，然后看看有什么技巧，看看它支持什么，然后你有时间回去。

09:29
We treat this stuff. Okay, so this is a bit different from the usual historical where usually you are working incrementally. You look at one set, it adds on the argument. You look at another set, it adds on in our argument.
我们处理这些东西。好了，这和以往的情况有点不同通常情况下你是增量式的。你看一组，它增加了论点。你看另一组，它增加了我们的论证。

09:40
And then towards the end, you sort of reach something that's called 95 % there, you might get a little bit, but it's always cumulative, right?
在接近尾声的时候，你达到了所谓的95%，你可能会得到一点，但它总是累积的，对吧?

09:47
Okay.
好吧。

09:48
If you go forward, and then if you don't push anymore, you wouldn't know they will break.
如果你往前走，如果你不再用力，你就不会知道它们会断裂。

09:55
Right until you try. So it's kind of like go from zero to 50 to 99, and then it bricks, and you go back to 70, and you go back forward again.
直到你尝试。这有点像从0到50再到99，然后它卡住了，然后你回到70，然后再向前走。

10:05
Right? So it's not accumulated, but that's something you need to expect it comes to if you do it.
对吧?所以它不是积累起来的，但如果你做了，你需要期待它的到来。

10:11
Okay.

好吧。

10:14
Okay.
好吧。

10:14
This is why this semester is like. This is to help you sort of push through all the way to the end with the project.
这就是为什么这个学期就像。这是为了帮助你完成整个项目。

10:23
All right, cool.
好吧，酷。

10:25
Okay. So these are our three sort of ambitions. Today, we're going to look and consider when AI might be a tool or work and when it's a method.
好吧。这就是我们的三个抱负。今天，我们将看看并考虑人工智能何时可能成为一种工具或工作，何时可能成为一种方法。

10:37
All right. And this two method.
好吧。这两个方法。

10:39
Kind of typology. It's very helpful for us to think a little bit about how we would approach and think about here, gen AI. when I say I today, I'm talking about the colors that we have. LS.
有点类型学。这对我们思考如何处理和思考人工智能很有帮助。当我今天说“我”的时候，我指的是我们拥有的颜色。LS。

10:52
Talk a little bit about what distinguishes that from computer programs and algorithms. That as long as this.

谈谈它与计算机程序和算法的区别。那个和这个一样长。

11:01
We look at what verification or responsibility we have as historians. So where humans and historians sit, we're doing this.
我们要看看作为历史学家我们有什么证据或责任。所以，无论人类和历史学家坐在哪里，我们都在做这件事。

11:13
Process, right? Of using AI and kind of thinking a wider view, but then thinking about sort of digital history, data sets themselves as.
过程中,对吧?使用人工智能，从更广阔的角度思考，然后想想数字历史，数据集本身。

11:23
Interests. Because when funded literature, we talk about matrix, we talk about data sets, and they're very, very similar in the way that we approach.
的利益。因为当被资助的文献，我们谈论矩阵，我们谈论数据集，它们在我们处理的方式上非常非常相似。

11:33
For the days of history and digital history, because you're essentially working with the same structure, same structure there.
对于历史和数字历史来说，因为你基本上是在用同样的结构工作，同样的结构。

11:40
Okay, we'll do a little bit of comparison, as the last one to know today.
好的，我们会做一点比较，作为今天最后一个知道的。

11:45
About the basic stories you think, right? The historians are the interdicators in terms of contexts, in terms of curation.
关于基本的故事，对吧？历史学家是背景和管理方面的仲裁者。

11:53
Instruction. Okay. So this is where we can go.
指令。好吧。这就是我们可以去的地方。

11:55
This is the focus questions, right? And, I mean.
这是重点问题，对吧？我是说。

12:02
Slightly generic at this point, but it's really to get you to think a little bit about how you might incorporate it within your own work, right?
在这一点上有点笼统，但它真的是让你思考一下如何将它融入到你自己的工作中，对吧?

12:09
How the readings are describing some of these processes and methods, right? And kind of ask me to explore a little bit. How might you apply it? How might you be seeing some of these.
阅读材料是如何描述这些过程和方法的？让我再探索一下。你会如何应用它呢？你是怎么看到这些的。

12:20
Concepts, right? So especially when they are key studies. When I give you key studies to read, they're not perfect. You will see that they're always errors. They have missteps along the Way.
概念,对吧?尤其是当它们是关键研究的时候。当我给你们一些关键的研究来阅读时，它们并不完美。你会发现它们总是错误的。他们在前进的道路上也有失误。

12:29
And you'll find that's a very common story.
你会发现这是一个很常见的故事。

12:32
Okay, so let's start from the very basics.

好吧，让我们从最基本的开始。

12:39
How does gen AI look? Does anyone, can anyone tell me or describe me how you think gen AI would It doesn't have to correct. That's what we're here for, right? But how do you think it works? What is an LM?
gen AI是什么样子？有人能告诉我或者描述一下你认为gen AI会怎样吗？这就是我们来的目的，对吧？但你认为它是如何运作的呢？LM是什么？

13:00
If you know the working.
如果你知道怎么工作。

13:08
What does it do?
它是做什么的?

13:09
And then they generate a pattern from the language. We look at the language. So when you say, look at the language, what do you mean specifically, e.g., if we're asking something from AI, and they see a pattern like, what is like, the context of what we Actually.
然后他们从语言中生成一个模式。我们看语言。所以当你说，看看语言，你的具体意思是什么，例如，如果我们向人工智能询问一些事情，他们看到一个模式，比如，我们实际上是什么样子的，上下文。

13:28
Want to find it. And then, in general, what they are programmed to find from the context and language that was used? Does anyone know where this context?
我想找到它。然后，总的来说，他们被设定从语境和语言中找到什么？有人知道这个上下文在哪里吗?

13:39
When you ask, when you ask a Gen Ai and LM, right, what is the weather

like in Singapore?
当你问Gen Ai和LM，新加坡的天气怎么样?

13:46
What is the process that happens behind this?
这背后发生的过程是什么?

13:49
If you're using a different source AI, it would like to search that website with whether the.
如果你使用不同来源的人工智能，它会搜索该网站是否。

13:59
That's using kind of.
这是在用。

14:11
Oh, for the step reasoning process, what's the basis of that? How does it know what to do?
对于步进推理过程，它的基础是什么？它怎么知道该做什么?

14:18
Even when you see enough of the similar sentences, they can simply predict that this is what the user wants.
即使你看到足够多的类似句子，它们也能简单地预测出这就是用户想要的。

14:32
So the LM is a guesser. LM is a.
LM是一个猜测者。LM是a。

14:37
Auto correct. It's a very, very advanced auto correct, a very, very advanced word suggestion. So, you know, when you type on your phone a message, I am going to the.
自动纠正。这是一个非常非常高级的自动纠错，一个非常非常高级的单词建

议。所以，你知道，当你在手机上输入一条信息时，我要去。

14:48
And then it will suggest several things to the canteen, to the cinema, to the classroom, to the whatever, right? And so it's trying to suggest to you what the next word.
然后它会给食堂，电影院，教室，等等提供一些建议，对吧？所以它试图告诉你下一个单词是什么。

14:59
Would be to complete a sentence. Okay.
将是完成一个句子。好吧。

15:02
Um.
嗯。

15:03
And so this is a right. They are essentially other countries. They are statistical models that predict very well nowadays what words or phrases or symbols come at the end of.
这是对的。他们本质上是其他国家。它们是统计模型，现在可以很好地预测单词、短语或符号的结尾。

15:17
What are the texts we're looking for? So E.G., when I say, what is the weather in Singapore, right? And then what are the words that come after this set?
我们要找的短信是什么？例如，当我说新加坡的天气如何时，对吧？然后这个集合后面的单词是什么？

15:29
Of words. What is the weather right in Singapore? So that's what is the weather?
的单词。新加坡的天气怎么样？这就是天气？

15:36
Six words, right? With this six words is a very specific configuration of words, right?
六个字，对吧？这六个单词是一个非常特殊的单词配置，对吧？

15:42
What words in the words corpus of information and data and examples of the English language, what words will come up to that? What are the most common?
在英语的信息和数据的语料库和例子中，哪些单词会出现在那里？最常见的是什么？

15:53
We will have a look at this more clearly. All right. But the general concept is, LMS are predictless, right? They are probabilistic.
我们会看得更清楚的。好吧。但总的概念是，LMS是不可预测的，对吧？它们是概率性的。

16:05
So they're looking for what is most common, what probably is what you're looking for.
所以他们在寻找最常见的，可能是你在寻找的。

16:09
They're trained to simulate texts and the sources, usually in a very specific context, because they're drawn from data that exists, that're trainer data that exists, right? So when we ask, what is the weather in Singapore, right?
它们被训练来模拟文本和来源，通常是在一个非常特定的环境中，因为它们是从现有的数据中提取出来的，这是现有的训练器数据，对吧？所以当我们问，新加坡的天气怎么样？

16:24
It will go to its data set, right? And it will look and evaluate all the languages in there, all the examples of sentences in there.

它会回到它的数据集，对吧？它会查找并计算所有的语言，所有的句子例子。

16:33
And these sentences come from this, like, like, second exchange Internet sources, right?
这些句子来自于，像，像，第二次交换网络资源，对吧？

16:41
What's the problem there?
有什么问题吗？

16:44
It's all existing knowledge. It's not really creating new knowledge. So.
这都是现有的知识。它并没有真正创造新的知识。所以。

16:54
I will just give the medium.
我只放一半。

16:56
So it tends to, it's the most probable. It tends to. It's the media, right?
所以它趋向于，它是最可能的。它倾向于。是媒体，对吧？

17:02
So the more details are from, kind of, the more targeted, you know, until.
所以细节越多，就越有针对性，直到。

17:08
So something can be very important, quite specific about what you're looking for, right? What's another problem?
所以有些东西可能非常重要，对你要找的东西非常具体，对吧？还有什么问题？

17:14

Think about maybe last one.
想想最后一个。

17:24
By talking about in general? Or is the conflict? Because right now the Constitution is like.
通过谈论一般？还是冲突？因为现在宪法就像。

17:40
Corpus is too general, so you may not be quite exactly what specialized questions may ask. So if your corpus is two of those general level statements or information that is.
语料库太笼统了，所以你可能不太清楚专门的问题可能会问什么。所以如果你的语料库是两个一般级别的语句或信息。

17:53
Pretty like low level, like, it's not very specific, not very specialized. You're not really going to get the kind of information that you.
很低级，不是很具体，不是很专业。你不会得到你想要的那种信息。

18:00
One, right? But it's general, in one way, by school specific, in other ways, in what ways are l M quas specific?
一个,对吧?但它是一般的，在某种程度上，具体到学校，在其他方面，在哪些方面M是特定的?

18:07
Put a between on.
戴上夹板。

18:12
English language causations online.
英语因果关系在线。

18:21

English languages, combinations online are be fully respective of human society.
英语语言、网络组合是人类社会的充分体现。

18:31
You know, first of all, there's a language barrier, right? There's a language bias in LMS because of the source either.
你知道，首先，有语言障碍，对吧？在LMS中，由于来源的原因也存在语言偏差。

18:40
There is a very specific socioeconomic viewpoint that is replicated in our own right. People for online to discuss, online and having conversations.
有一个非常具体的社会经济观点在我们自己的权利中得到了复制。人们在网上讨论，在网上进行对话。

18:53
Forms online, right from very specific periods, use specific types of words and phrases have very specific symbols.
在线表格，从非常特定的时期开始，使用特定类型的单词和短语，具有非常特定的符号。

19:03
Shorthands that they're using when they refer to, when they say the word monster, they have very specific cultures, and the word monster is right. So let's say, what is the monster that you're from.
当他们说到“怪物”这个词时，他们使用的速记，他们有非常特殊的文化，“怪物”这个词是对的。比方说，你来自哪个怪物。

19:14
Right? What are they going to feedback to you? They're going to feedback to you the data that they train up, right? Which maybe for them.
对吧?他们会给你什么反馈？他们会把训练出来的数据反馈给你，对吧？也许对他们来说是这样。

19:20
Dragons and monster. But in these Asian contexts, and dragons saw the monster.
龙和怪物。但在这些亚洲背景下，龙看到了怪物。

19:24
Right? So thinking about how LM to train the source of the data.
对吧?想想LM如何训练数据的来源。

19:33
It's really, really important, because you immediately can figure out where their bias is in life.
这真的非常非常重要，因为你可以立即发现他们在生活中的偏见。

19:39
Make sense.
讲得通。

19:43
Okay. How are we doing with these concepts?
好吧。我们是如何处理这些概念的?

19:50
It's a difficult.
这很难。

19:52
Moving a bit slowly today, because this one is a bit difficult and fundamental. Give me a read if you need to pass some useful something.
今天讲得有点慢，因为这个有点难，也有点基础。如果你需要传递一些有用的东西，给我读一下。

20:00
If not, I think you okay, cool.
如果没有，我觉得你没事，很好。

20:04
Do lls understand?
我明白了吗?

20:11
Reason like today, today, more words and people, but more.
道理就像今天，今天多了话和人，却多了。

20:18
Yes, yes, that's right. Yes. You are sometimes with the other.
是的，是的，没错。是。你有时和另一个人在一起。

20:28
Our answer is no. You don't.
我们的答案是否定的。你不。

20:32
Understand it in the way that you think we understand it, right? So let's use a human example. If I wanted to know the weather is.
用你认为我们理解它的方式去理解它，对吧？我们来举一个人类的例子。如果我想知道天气是。

20:41
Singapore. What would I do? I would Google, right? Or Google? I look for a source of information that, as I saw, an ID reliable.
新加坡。我该怎么办？我会谷歌，对吧？还是谷歌?我寻找的信息来源，正如我所看到的，是一个可靠的身份。

20:53
Right? I'll go to the method list and see E.G., the analysis, and I'll be like, okay, so what is the historical climate information there? What's the graph in the data?
对吧?我会去到方法列表，看看例如，分析，我会说，好的，那么历史气候信息是什么？数据中的图形是什么？

21:02
Today, whether it's going to rain this month or next month or whatever, right? So this is what I do. I go to the library and I pick up the right book, or the right source of.
今天，不管这个月下雨还是下个月下雨，对吧？这就是我所做的。我去图书馆，我拿起正确的书，或者正确的来源。

21:13
You need information, right? I'll answer that. You don't go to a specific source. Look information.
你需要信息，对吧？我来回答这个问题。你不用去找一个特定的来源。看信息。

21:20
At the most basic level, you don't do that.
在最基本的层面上，你不会这么做。

21:24
They are taking all the sensors, they are stripped up from all the different sources. They just pout them to get.
他们把所有的传感器，从所有不同的来源中剥离出来。他们只是撅着嘴。

21:30
And then from there, they're just, oh, okay, we will define from there the sense of the weather in Singapore is.
然后，他们就会说，哦，好吧，我们将从那里定义新加坡的天气。

21:37
Cloudy 1 million times, sunny, 2 million times, wet, 0.5 million times. So probably it will be sunnyva, because sunny is 2 million. Not great.
阴天100万次，晴天200万次，阴天50万次。所以可能是sunnyva，因为sunny是200万。不伟大。

21:49

I think the most, the one that is the most prominent, the most probability you probably be the right answer.
我认为最突出的，最有可能是正确答案的。

22:00
Okay.
好吧。

22:00
I don't understand, necessarily. We're just looking at all the rules. Let's keep saying.
我不一定理解。我们只是在看所有的规则。我们继续说。

22:08
So this is what we call the tocanaser. The Tocanaser is what we've been doing right. What is the weather in Singapore is six weeks.
这就是我们所说的tocanaser。Tocanaser就是我们一直在做的事情。新加坡的天气是六个星期。

22:21
Right? Let's say possibility. Say each word is a token. What is two tokens? The, what is the three tokens?
对吧?我们说可能性。说每一个字都是一个记号。两个代币是什么？这三个记号是什么?

22:31
So finding the two tokens, finding the right three tokens, finding the right balance, both of them so and so forth, right? So let's say we only give them this four.
所以找到两个代币，找到合适的三个代币，找到合适的平衡，等等，对吧?假设我们只给它们这个4。

22:42
Then they will try and fill in what the rest is.
然后他们会尝试填入剩下的部分。

22:45
And what they do is you look at finding the, finding the right finding the right balance, and then they look at the right, the right balance.
他们所做的就是你要找到正确的找到正确的平衡，然后他们要找到正确的平衡。

22:56
Right? You'll compare all these tokens against each other. You look at the order of the tokens, right? And then you'll decide, okay, so when tokens say token number one and token number doing.
对吧?你将把所有这些令牌相互比较。你看一下符号的顺序，对吧？然后你会决定，好吧，所以当令牌说令牌1和令牌数做。

23:06
Number three. I can hear that the most likely to open that will come out of third is to number four.
3号。我听说最有可能从第三位打开的是第4位。

23:11
This is what I give you. It. Predict.
这是我给你的。它。预测。

23:14
Yes, I think from my essay, it's a bit ironic, because in a way, this seems very like different from how we process information, but in a way, it's also based on the neural network, and certain, like weights given to Yes, so it's like.
是的，我认为从我的文章中，这有点讽刺，因为在某种程度上，这看起来和我们处理信息的方式很不一样，但在某种程度上，它也是基于神经网络的，而且肯定的，就像给Yes的权重，所以它就像。

23:30
It seems very different from us. But in a way, it's such an advancement in the air. It's only possible because they refugees.

它看起来和我们很不一样。但在某种程度上，这是空中的一大进步。这是可能的，因为他们是难民。

23:37
That's right.
这是正确的。

23:39
So we'll talk a little bit about weeks later on, right?
我们几周后再谈，对吧?

23:43
What this means, essentially, what's going on here is that.
这意味着，本质上，这里发生的是。

23:46
Let's use the Singapore weather example again. Okay. What is the weather in Singapore? You have write this on.
让我们再次以新加坡的天气为例。好吧。新加坡的天气怎么样？你把这个写在上面了。

23:56
Good. What's it? Um.
好。它是什么?嗯。

24:10
It was got sunny.
天气晴朗。

24:22
About, let's be familiar.
关于，让我们熟悉一下。

24:25
What else? You just know what?

还有什么？你知道吗?

24:29
Which was.
这是。

24:33
Right? And.
对吧?和。

24:37
Whatever doesn't matter. Okay, two, right. Cool.
无所谓了。好的，两个。酷。

24:45
So this is what we credit, right, that I would credit, because 2 million businesses are open five So this is probably the right answer.
这是我们的信用额度，对吧，我会记入信用额度，因为有200万家企业在营业，所以这可能是正确的答案。

24:56
Okay. Now there's something else also going behind the scene, because the word weather.
好吧。这背后还有别的东西，因为天气这个词。

25:02
Has a particular context, right? But we understand whether it's like climate, like the actual physical weather, right? But what if we are thinking about in more specialized senses as well?
有特定的背景，对吧？但我们知道它是否像气候，像实际的物理天气，对吧？但如果我们从更专业的角度来思考呢?

25:13
In the most special license. What is this? Where is metaphorical?
在最特别的执照上。这是什么？隐喻在哪里?

25:17
What if it's like the tempest within, right stone inside of art? The psychological, metaphorical reading of weather?
如果这就像内心的暴风雨，对吧，艺术内部的石头？对天气的心理隐喻解读?

25:26
Right? So think about it like this. There's a, what we call sort of a cloud of contexts, right? Whether it's associated with things like Sammy, where?
对吧?所以这样想。有一种我们称之为上下文云的东西，对吧？它是否与萨米之类的东西有关，在哪里?

25:38
Windy, cloudy, right? All these are physical manifestations Celsius, fahrenheit. These are all goods. They are very closely related, right? They are co, located.
多风，多云，对吧？所有这些都是物理表现摄氏度，华氏度。这些都是商品。它们是密切相关的，对吧？他们是co，定位。

25:48
Found around this concept of weather. But then there is also another way that we use to work weather.
围绕着天气这个概念。但是我们还有另一种方法来计算天气。

25:59
Right?
对吧?

25:59
But let's see, um.
但是让我想想。

26:03
Psychological, right? So, like, um, have.

心理,对吧?所以，嗯，有。

26:10
Post, tragic.
帖子,悲剧。

26:14
Sad, right? Happy, right? So these are emotional, psychological, right? And sometimes it includes.
难过的时候,对吧?开心,对吧?这些是情感上的，心理上的，对吧？有时它包括。

26:26
Weather. But at the same time, you also have another context cloud, right where it's talking about these physical phenomena.
天气。但与此同时，你也有另一个上下文云，就在它谈论这些物理现象的地方。

26:35
So because we have billions and billions of examples of language, the LM can actually figure out whether you're using, whether.
因为我们有数十亿的语言例子，LM可以知道你是否在使用，是否。

26:46
In terms of the psychological context, or whether it's in the physical context, right? Account two, at least two parts can go when we say whether here, when this person puts in the word, whether doing the psychological way.
无论是心理环境，还是生理环境，对吧？解释二，当我们说，当这个人说这个词的时候，至少有两个部分是可以的。

26:59
Possible weather.
可能天气。

27:02
So it kind of makes a decision. That decision is what we call that. It's just the new analysis, the weight. It's more likely, usually.
所以它做了一个决定。这个决定就是我们所说的。只是新的分析，权重。通常更有可能。

27:13
That whether it refers to physical stuff, maybe there's like a 10 % chance that it's referring to psychological.
不管它指的是物理上的东西，也许有10%的可能性指的是心理上的东西。

27:22
Right?
对吧?

27:22
And this white opens that important, because if the prompt is not weather in Singapore is but the prompt is the weather in cite me is, then they go, ah.
这个白色打开很重要，因为如果提示不是新加坡的天气而是cite me的天气，他们就会说，啊。

27:33
I think the word insight clues you that is this, rather than this.
我想“洞察力”这个词给你的提示是这个，而不是这个。

27:38
Good.
好。

27:38
Does that sort of make sense?
这样讲得通吗?

27:43

The algorithms are essentially parts of these contexts.
算法本质上是这些上下文的一部分。

27:54
And he's able to statistically run that to determine the context and then put in the.
他能够通过统计来确定上下文，然后放入。

28:00
The right to put on the right letter, yes. So that the LMSL is an algorithm, and it makes certain decisions. It's like a path. I'm going to walk down this path, and I'm going to say, okay, now I look at the word weather.
正确的字母，是的。所以LMSL是一种算法，它会做出某些决定。就像一条小路。我要沿着这条路走下去，我会说，好吧，现在我看看weather这个词。

28:13
Does this mean psychological weather, or this mean physical weather? If there's no information, it's probably physical weather. But if it says weather inside there, I'll be like, inside. What does that even mean.
这是指心理天气，还是指生理天气？如果没有信息，可能是物理天气。但如果里面写着天气，我就会说，在里面。这到底是什么意思。

28:23
Right? Whether insight probably means psychological side of what all this topic is.
对吧?洞察力是否意味着这个话题的心理层面。

28:27
Okay. And if you walk down this path, then you start using the context, the happy set, tragic and all that right, to reinterpret what's going on.
好吧。如果你沿着这条路走下去，你就会开始使用情境，快乐的场景，悲剧的场景等等，来重新解释正在发生的事情。

28:37
And that's why it sounds quite convincing. It sounds like it knows what

you're doing, because it's actually just mimicking what other people have said in this context.
这就是为什么它听起来很有说服力。听起来好像它知道你在做什么，因为它实际上只是在模仿别人在这种情况下说的话。

28:46
But doesn't actually understand, because it's all statistical, it's all probabilistic, right? So the more information you give it.
但实际上并不能理解，因为这都是统计性的，都是概率性的，对吧？你给它的信息越多。

28:55
The longer you're from, the more context it is given, and the better it is able to pick the right task.
你的时间越长，它所处的环境就越多，它就越能选择正确的任务。

29:03
That you want, that you're looking for.
你想要的，你在寻找的。

29:07
You give it more clues as to which are the right contexts they ought to be looking for.
你给了它更多的线索，告诉它哪些是它们应该寻找的正确语境。

29:14
If you get a very short comment forward, you'll have to just make a space after in case.
如果你得到一个很短的评论，你必须在后面留一个空格以防万一。

29:21
How are you doing? Very good. So, sorry. So is it token as an individual word?
你好吗？很好。所以,对不起。那么它是一个单独的单词吗?

29:32
Not always. I'll be a part of a word.
不总是正确的。我将成为一个世界的一部分。

29:36
The finding fine can be important.
这个发现可能很重要。

29:40
All right, but let's say, okay, so this is, this is an actual example of jgpts tokenize it, just to give you a sense.
好的，但是我们说，这是，这是一个jgpts标记它的实际例子，只是给你一个感觉。

29:49
What it is. The full stop here is actually a token. The G here is actually a token. The PT is another token.
它是什么。这里的句号实际上是一个记号。这里的G实际上是一个记号。PT是另一个令牌。

29:56
So in so forth, a token. This is what token. Here is a token. The nicer part is another token. Kind of look.
等等，一个记号。这是什么记号。这是代币。更好的部分是另一个令牌。有点像。

30:03
Poma. So even a punctuation gives information about which contexts.
Poma。所以即使是标点符号也能提供上下文的信息。

30:08
B or M, she perform.
B或M，她执行。

30:14

Okay.
好吧。

30:14
This is how we know when you start something wrong. Because let's say, I must start finding that F I N D I N G I.
这就是我们如何知道你什么时候开始做错了。因为我们说，我必须开始求出F I N I N G I。

30:25
To I flip the IMM, FM IB, IMG, right? The LM would be a bit not confused, but they will look at FMI.
翻转IMM， FM， IB， IMG，对吧？LM不会有点困惑，但他们会看FMI。

30:35
And I'll say, if there isn't really an instance where FFI really appears, and usually when FNI usually appears, it will be in the same context club as fi.
我会说，如果没有FFI真正出现的实例，通常当FNI通常出现时，它会和fi在同一个语境俱乐部。

30:45
It overlaps almost perfectly. And then that's when they know, you probably mean by them.
它几乎完美地重叠。然后他们就知道了，你说的他们可能是指他们。

30:51
Because the words around FMI are the same words, they are afim.
因为FMI周围的单词是相同的单词，所以它们是一致的。

30:58
Right?
对吧?

30:58
Makes sense.

是有意义的。

31:00
A good concept with your dish. I hopefully this gives you a good idea about how urbans work.
对你的菜有很好的概念。我希望这能给你一个关于城市如何运作的好主意。

31:09
I don't really explain the fact the difference between AI and LLM, not yet, but we made a distinction between computer programs and algorithms and gen AI.
我还没有真正解释人工智能和法学硕士之间的区别，但我们区分了计算机程序、算法和人工智能。

31:20
So gen AI and LM are kind of engineer.
所以gen AI和LM都是工程师。

31:25
Generative AI. Right? Generative AI. It generates outputs for you, right?
生成人工智能。对吧?生成人工智能。它为你生成输出，对吧?

31:34
Misleading, because all algorithms generate a result or an output anyway, but in this case, generated, they're thinking about, in a more creative sense, we're generating something that looks like it's created By.
误导，因为所有的算法都会产生一个结果或输出，但在这种情况下，生成，他们认为，在更有创造性的意义上，我们生成的东西看起来像是由。

31:46
And talent. Just, yes, actually, that's an interesting way for understanding AI, which is proposed by Invidia, which is like, that's a five layer key. So the way that it starts is like the.
和才华。是的，这是理解人工智能的一种有趣的方式，这是英伟达提出的，这是一个五层键。它开始的方式就像。

31:57
Energy and the infrastructure, or the chips, then, or chips and infrastructure, then the lms, and then, which is your general lms. Then you go up to the specialized LM. So that one is quite interesting to like.
能源和基础设施，或者芯片，或者芯片和基础设施，然后是lms，然后是一般的lms。然后再到专门的LM。所以这个很有趣。

32:10
I'm looking into if you're thinking about how to understand.
我想知道你是否在思考如何理解。

32:13
We'll talk a little bit about where this overlaps, because with research work, you have decisions to make, or whether you want to use GNAI slash LM.
我们会讨论一下这些重叠的地方，因为在研究工作中，你需要做出决定，或者你是否想要使用GNAI / LM。

32:29
Or whether you want to use specialized algorithms which have existed pre talent.
或者你是否想使用已经存在的专门算法。

32:34
In fact, all the tools that I teach you are pre genderi or very specific.
事实上，我教你们的所有工具都是非性别的或者非常具体的。

32:40
But lm's match language models, right with a cruising image. That's a model that is built out from.
但lm的语言模型与巡航图像相匹配。这个模型是建立在。

32:53
Much language practice.
多做语言练习。

32:54
Statistical model of how a language works, what content is in your language.
一门语言如何运作的统计模型，你的语言中有什么内容。

33:02
So predictor.
所以预测。

33:04
It doesn't actually know. No, it's just a very, very good question. From all the content and the problems and the working benefits with them, you will guess really well.
它实际上并不知道。不，这是个非常非常好的问题。从所有的内容、问题和使用它们的好处来看，你会猜得很好。

33:15
This is, this is in tact terms, is what we call certain numbers. We are using numbers to solve the problem.
这是，这是委婉的说法，这就是我们所说的特定数。我们正在用数字来解决这个问题。

33:26
The more numbers to exploit it, right? You bash, you literally bash the dollar with none.
利用它的数字越多，对吧？你真的是在无中生有地打击美元。

33:34
Others, right? You're trying to solve a problem by group, false.
其他人,对吗?你想要集体解决问题，错了。

33:37
The more data you have, the more people you are to attach to the problem.
你拥有的数据越多，你就需要更多的人来解决这个问题。

33:43
I think, think a bit like if you want to check someone's account.
我想，有点像你想检查某人的账户。

33:52
Or you want to unlock a PET lock, right? You could do it two ways. You could go from, let's say, a PET lock, two digits, zero, zero, zero, to nine, nine, Nine.
或者你想打开PET锁，对吧？有两种方法。你可以从，比方说，一个PET锁，两位数，0,0,0，到9,9,9。

34:02
You could try every single one until.
你可以尝试每一个，直到。

34:04
Works. Or you can be like, what is a three digit number that's significant for this person?
作品。或者你可以说，哪个三位数对这个人有意义?

34:10
Birthday, right? Okay. Now about this. Two, nine, eight or eight. Two, nine. Let's try this, right. So that's the reading process where you're trying to access.
的生日,对吧?好吧。现在说说这个。二、九、八、八。2、9。我们来试试这个。这就是你试图进入的阅读过程。

34:21
Accents, trying to understand the person's reason. And Alan doesn't do that. I am the same. She's like a train everything. I've had it right.
口音，试着理解对方的原因。艾伦不这么做。我也一样。她就像一列火车。我猜对了。

34:30

Now, the result here, the consequence here.
现在，结果在这里，结果在这里。

34:35
It's of course, the more words you give it, the better the prediction will be like, right? The more details you give it about what you're looking for, the better the output will be.
当然，你给它的单词越多，预测就越准确，对吧？你提供的信息越多，输出的结果就越好。

34:44
The problem is it's very computationally expensive.
问题是它的计算成本很高。

34:47
The more tokens you have, the more it has to do the comparisons between each part of the form.
您拥有的令牌越多，它就越需要在表单的每个部分之间进行比较。

34:53
Right? So remember, just one, we said, fourth opens, we'll do one versus this one versus three. One versus four.
对吧?记住，只有一个，我们说过，第四个开口，我们做一个对这个对三个。一对四。

34:59
And then one, two, three, four, one and two. This is three, one and two, versus three and four.
然后一，二，三，四，一，二。这是三，一，二，而不是三，四。

35:05
So and so forth. So they're doing these comparisons with people things, right? You can imagine it escalates a lot exponentially as you add more words, and then.
诸如此类。所以他们在和人做比较，对吧？你可以想象，随着你添加更多的

单词，它会呈指数级增长，然后。

35:14
This is why it's computationally expensive, right? The longer the problem, the more has And every time you give a prompt, it gives you a result, right? And you add in another prompt. What does it do?
这就是为什么它在计算上很昂贵，对吧？问题越长，问题越多每次你给出提示，它就会给出一个结果，对吧？然后添加另一个提示符。它是做什么的?

35:26
It doesn't actually follow on for the same thing. It goes back and looks at everything that has come before your original prompt, the original response and your new prompt. It takes all of that, and it reevaluates the whole thing again.
它实际上并不是对同一件事的延续。它会返回并查看在原始提示，原始响应和新提示之前的所有内容。它把这些都取走，然后重新求值。

35:39
Right?
对吧?

35:39
This is why you can regenerate your response, because you're just taking the entire thing as your prompt again.
这就是为什么你可以重新生成你的响应，因为你只是把整个事情作为你的提示。

35:49
And you valuating everything that was it, whether it's your original form or what was generated for you.
然后计算所有的东西，无论是原始形式还是生成的形式。

35:55
It's a very expensive competition.
这是一个非常昂贵的比赛。

35:59
Yes, I have a question. What is the editing image or?
是的，我有一个问题。编辑的图像是什么？

36:08
It's a good question. How does a machine read? Image?
这是个好问题。机器是如何阅读的？形象吗？

36:15
Right? We can get a little bit into this, usually, as we know.
对吧?我们可以稍微深入一点，通常，我们知道。

36:22
These are the weeks that we're talking about this time.
我们这次讨论的是这几周。

36:28
Okay, so let's say we have an image.
假设我们有一个图像。

36:31
Of an animal, falling animal, and we're trying to figure out whether it's a cat or a dog, a machine's trying to understand whether it's a cat.
一只动物，一只坠落的动物，我们试图弄清楚它是猫还是狗，一台机器试图理解它是否是猫。

36:41
So a machine will be like, okay, so we'll pick every single pixel of image. Remember, pixels right of an image. An image is a, what a vector or rasper?
所以机器会说，好吧，我们会选择图像的每一个像素。记住，像素在图像的右边。图像是矢量还是栅格？

36:51

Arrester, right? Is a great of colors, right? And so what a machine does is going to take every single person.
避雷器,对吧?是一个伟大的颜色，对吧？所以机器要做的是把每一个人都带走。

37:00
Right? And then he's going to ask second questions.
对吧?然后他会问第二个问题。

37:03
Based on the region of the pixels, is this align, right? If you've got a lot white, and then you've got a lot black or dark, then that corner.
基于像素的区域，这是对齐的吗？如果你有很多白色，然后你有很多黑色或深色，然后那个角落。

37:15
Comes probably a lot, right? Or is it a circle, right? Does the thing art this way, or does it actually go wrong like that?
可能会来很多次，对吧？或者它是一个圆，对吧？这个东西是这样设计的，还是会像那样出错?

37:24
So is the pixels when you put together? Is that a circle, or is it a line? Right? So, basic stuff, we are decomposing a cat into lines and circles.
那么当你把像素放在一起的时候呢？这是一个圆，还是一条线？对吧?所以，基本的东西，我们把猫分解成线和圆。

37:33
So that we can ask, is this a triangle? Because the cat Years got triangles right?
所以我们可以问，这是个三角形吗？因为猫年画的三角形对吗?

37:39
Angle in the image.
图像中的角度。

37:41
Is this for? Right? So progressively moving from basic shapes, lines, circles, triangles, into stuff that is a bit more.
这是什么？对吧?所以逐渐从基本的形状，直线，圆，三角形，到更。

37:52
Complex, right? A year for I notes, right. So what is a nose? Two circles may be surrounded by a larger or oil.
复杂的,对吧?1美元一年，对吧。那么什么是鼻子呢？两个圆圈可能被一个较大的或油包围。

38:04
Avoid shape, right? So if you have two circles and near to each other, and then there's a oval that surrounds them that is probably a nose.
避免形状，对吧？所以如果你有两个圆圈彼此靠近，然后有一个椭圆形围绕着它们那可能是鼻子。

38:12
If you have two triangles, they're roughly knit each other, factors each other approximately.
如果你有两个三角形，它们大致是相互交织的，互相近似因式分解。

38:19
Those are probably years, right?
那可能是几年，对吧?

38:23
If you have sort of an undefined outline, right? That's probably four.
如果你有一个未定义的大纲，对吧？可能是4。

38:29
Right? So the machine recognizes these basic shapes is, has been taught and trained that these like two psychotics to each other and one over around it.

对吧?机器可以识别这些基本的形状，被教导和训练它们就像两个精神病患者，一个在它周围。

38:42
That's a nose. A cat must have a nose. So, okay, it could be a cat. If it doesn't have a nose, then probably not a cat.
那是鼻子。猫必须有鼻子。好吧，可能是只猫。如果它没有鼻子，那可能不是猫。

38:50
Right?
对吧?

38:52
So let's say, let's assume that only cats have triangular use, and dogs don't have only appropriate use, right? So you look at this picture of dog.
我们假设只有猫有三角用途，而狗不只有适当用途，对吧？你看这张狗的照片。

39:02
Everybody, oh, we have a nose, you have a eye. But they're no triangles.
大家注意，我们有鼻子，你们有眼睛。但它们不是三角形。

39:06
Hopefully not, I guess.
希望不会，我想。

39:08
So this is finished classification. Okay.
这就是完成的分类。好吧。

39:12
Now to teach an algorithm that a cat equates in two triangles, several circles, so on and so forth. This is.
现在教一个算法，一只猫等于两个三角形，几个圆，等等。这是。

39:20
Where we come in. As soon as we tell a computer, this is cat. This is cat. This is a dog. This is a dog. This is a dog. That's a cat. So and so forth. And then it kind of.
这就是我们的切入点。只要我们告诉电脑，这是猫。这是猫。这是一只狗。这是一只狗。这是一只狗。那是一只猫。诸如此类。然后它有点。

39:30
Learns. When you say it learns, it means it will go half, okay, two circles. And an oval needs a nose. So.
学习。当你说它学习的时候，它的意思是它会走一半，好吧，两个圈。椭圆形的人需要鼻子。所以。

39:39
Circle, circle, circle, google put together notes.
圈，圈，圈，谷歌把音符放在一起。

39:43
Right? So this is the neural network that might be talking about, right? These are the ways we train a computer to follow certain lines of reasoning, the same way that we call it a neural network, because this is how electrical impulses and how we move from one neuron to another.
对吧?这就是我们要讨论的神经网络，对吧？这就是我们训练计算机遵循特定推理路线的方法，我们称之为神经网络的方法，因为这就是电脉冲和我们如何从一个神经元移动到另一个神经元的方式。

39:58
And the firing of the path, the path of that electrical impulse, is what we think is probably kind of reasoning.
电脉冲的发射路径，电脉冲的路径，我们认为可能是一种推理。

40:08
We don't know yet.
我们还不知道。

40:09
And because it's so complex, we actually don't, we cannot trace the past exactly, even in.
因为它是如此复杂，我们实际上不能，我们不能准确地追溯过去，即使在。

40:18
The algorithms, we can.
算法，我们可以。

40:20
Can get a sense why we call weeks, because let's say two triangles, right? Like that. So goes like that. Below it, 70 % will be a cat.
可以理解为什么我们称周为周，因为我们说两个三角形，对吧？像这样。就像这样。在它下面，70%是猫。

40:30
But probabilistic, right? The deeper think of something else, right?
但是概率，对吧？越往深处想别的东西，对吧?

40:37
So these are all weeks. We are telling it, this copy a cat, but this is a cat. We tell it is a cat, right? And so we are weeping it. We're weeping this configuration a bit more.
这些都是周数。我们告诉它，这个拷贝是一只猫，但这是一只猫。我们说它是一只猫，对吧？所以我们在哭泣。我们再看一下这个构型。

40:49
We're saying, this is cat.
我们说，这是猫。

40:50
These are why LMS are different, because they read the different meanings.
这就是LMS不同的原因，因为它们读取不同的含义。

40:57
Right? These are why LMS have personalities, some of personalities.
对吧?这就是为什么LMS有个性，一些个性。

41:02
Because we've weakened it in the popular way. We've trained it in different ways. Different companies trained our owns in different ways. The basic concepts of the same architecture is the same.
因为我们用流行的方式削弱了它。我们用不同的方法训练它。不同的公司以不同的方式培训我们的员工。相同架构的基本概念是相同的。

41:12
Right? They're all elements. But at some point there's a training where humans are sitting there and saying, this is a useful response. No, this is not detailed enough.
对吧?它们都是元素。但在某种程度上，有一个训练，人类坐在那里说，这是一个有用的反应。不，这还不够详细。

41:23
You need more detail for this. No, we prefer bullet points rather than three sentences.
你需要更多的细节。不，我们更喜欢要点而不是三句话。

41:28
Right? So there's why chechen here is very business, right? This is a summary. Here's a bullet point, bullet point, bullet point bullet plot is very luxurious.
对吧?这就是为什么这里的车臣很商业化，对吧？这是一个总结。这里有一个点,点,点子弹情节非常豪华。

41:38
Very, lots of pros, long sentences, a lot of contexts, and always ends up with, what do you want to look at?
非常非常多的优点，长句子，很多上下文，最后总是说，你想看什么?

41:44
More now, right? It's very interrogative. You can see, after CEO is a bit correct. Literature Graduate.
现在更多了，对吧？这是疑问句。你可以看到，CEO之后是有点正确的。文学研究生。

41:51
So they're interested in building a more humanistic AI, and whether that's a contradiction in terms of the difference.
所以他们感兴趣的是建立一个更人性化的人工智能，而这在差异方面是否存在矛盾。

42:00
Okay. All right. How are we doing? We just covered a lot of ground. Yes, course. Are we good?
好吧。好吧。我们做得怎么样？我们刚刚走了很多路。是的,当然。我们还好吗?

42:11
Just some working knowledge for you to get a sense of what, okay? You have a sense of what is Google and what's considered.
只是一些实用的知识让你了解一下，好吗？你知道什么是谷歌，什么是被考虑的。

42:19
The black boxing will come back to us again.
黑色拳击将再次回到我们身边。

42:22
All right. Something that I didn't mention here is that this is kind of, this is an algorithm.
好吧。我在这里没有提到的是，这是一种算法。

42:31

So this happens within the black box of the children, right? And what it does is it gives us a determination whether it thinks this picture is a cat or dog.
所以这发生在孩子们的黑盒子里，对吧？它所做的就是告诉我们它认为这张照片是猫还是狗。

42:42
No, sorry, whether is a cat or not, right? So, yes or no, this is a cat, yes or no, right? So it gives us the result. And then we as humans.
不，抱歉，不管是不是猫，对吧？所以，是或不是，这是一只猫，是或不是，对吧？它给出了结果。然后我们作为人类。

42:51
Here in this process, where we go, we will write.
在这个过程中，我们走到哪里，我们就写到哪里。

42:56
Or you're wrong. And then every time we give it more information, every time we tell it you are right, or every time we tell it you are wrong, it learns by Adapting.
或者你错了。然后每次我们给它更多的信息，每次我们告诉它你是对的，或者每次我们告诉它你是错的，它都会通过适应来学习。

43:07
The process here, by changing the weights here, right? Maybe we shouldn't wait two triangles. So importantly, in determining whether it's a cat.
这个过程，通过改变这里的权值，对吧？也许我们不应该等两个三角形。所以重要的是，判断它是否是一只猫。

43:16
Because sometimes cats turn on one side, right? So when there's a force for when the force package is right, there was only one triangle because only one year was there.
因为有时候猫会朝一边转，对吧？所以当有一个力当力包是正确的，只有一

个三角形因为只有一年。

43:26
But humans, we know it's a cat, right? But the computer says, there's only one triangle, there's not two triangles, so it's definitely not a cat. And then we tell it's wrong, then we go, oh, maybe I shouldn't have been so confident.
但是人类，我们知道这是一只猫，对吧？但是电脑说，只有一个三角形，没有两个三角形，所以它肯定不是一只猫。然后我们说这是错的，然后我们想，哦，也许我不应该这么自信。

43:37
That two triangles would definitely mean cat. We must have two triangles. Maybe one triangle is also a cat.
那两个三角形肯定代表猫。我们必须有两个三角形。也许其中一个三角形也是一只猫。

43:43
This is the correction mechanism. We are correcting it, we're telling it, and then we're refining this so that they give us, kind of the results that we.
这就是校正机制。我们在纠正它，我们在告诉它，然后我们在改进它，这样他们就会给我们，某种我们。

43:53
One, or that we would have done others.
一个，或者我们会做其他的。

43:56
That's why this is a treat secret. The weeks are treat secrets.
这就是为什么这是款待的秘密。这几周是秘密之周。

44:04
These are not publicly available. So GPD has its own wits. Claw has its own wits. All these things have done wits, and it's still a secret source.
这些都不是公开的。所以GPD有自己的智慧。Claw有它自己的智慧。所有这

些东西都很聪明，它仍然是一个秘密来源。

44:15
This is why, if you're using now, and you are happy for them to train on you, you're contributing back to the caucus, and then they will use.
这就是为什么，如果你现在正在使用，并且你很高兴他们对你进行培训，那么你就为核心小组做出了贡献，然后他们就会使用。

44:25
To further develop their model and train their reads.
进一步发展他们的模式，训练他们的阅读能力。

44:28
Okay. All right. Any questions.
好吧。好吧。任何问题。

44:36
Any questions.
任何问题。

44:38
All right. So, um.
好吧。所以,嗯。

44:46
So we're talking about tokens, right? We're talking about the fact that tokens means. The way that it works is that there are so many.
我们讨论的是代币，对吧？我们讨论的是令牌的意义。它的运作方式是有很多。

44:56
Computational steps, computations and calculations you need to make. The longer you're from the.
计算步骤，你需要做的计算和计算。你离得越远。

45:05
Longer the response comes out, more populations need to be done.
反应出来的时间越长，需要做的人口就越多。

45:08
Um and select in terms of billions of.
以数十亿计来选择。

45:11
Populations, so they're very expensive.
人口，所以它们非常昂贵。

45:14
Require electricity. Require water to cool. This is why.
需要电力。需要水来冷却。这就是原因。

45:23
Providers will cut off your request after a while. Right? I think let's see Gemini is at 1 million province at the moment.
供应商会在一段时间后切断您的请求。对吧?我想让我们看看双子座现在是100万省。

45:34
1002000, something. Clock five is that a million? Plot, 4.6 % 400000.
1002000,什么。5号钟是100万吗？情节，4.6% 400000。

45:42
Right? And we will see it open to mean the entire thing, including your passport, free of the chat.
对吧?我们会看到它打开意味着所有的东西，包括你的护照，没有聊天。

45:48
So at some point you'll say, too much, too expensive, right? It's not a new chat.
所以在某种程度上你会说，太贵了，对吧？这不是一个新话题。

45:55
This way, you run up against.
这样，你就会遇到。

45:57
Visa limitation.
签证限制。

46:00
Um.
嗯。

46:03
The reason why I don't really focus on Chen Ai in this course.
我在这门课上不太关注陈艾的原因。

46:08
Just because it's not free, there's a lot of tools that are available to everyone.
正因为它不是免费的，所以每个人都可以使用很多工具。

46:13
If you are willing to pay more, you are able to access a better tool, essentially.
如果你愿意花更多的钱，你就能获得更好的工具。

46:21
And it does run against some of the ethos that we have in digital humanities.
它确实违背了我们在数字人文学科中的一些精神。

46:26
Which is a very sharing equal. It's a very open source.
这是一个非常平等的分享。它是一个非常开放的源代码。

46:30
Right? Everyone should be able to replicate the same results using the same tools. So the stuff that you see in go Digital Scholarship Lab, right? The six tools, and there are tools that were treated by digital.
对吧?每个人都应该能够使用相同的工具复制相同的结果。你们在数字奖学金实验室看到的东西，对吧？这六种工具，有些工具被数字化处理过。

46:43
And it's in the nineties and the 2000s and early tens right. Four researchers, students got free. You can actually inspect the code in the recording unit.
这是在90年代，21世纪和10年代初。四名研究员，学生获得免费。实际上，您可以检查记录单元中的代码。

46:52
Right with LMS. You come, you never know. The weeks.
对LMS。你来了，你永远不会知道。这个星期。

46:57
And this was a probabilistic Every time you put in a prompt, you might get something slightly different.
这是一个概率每次你输入一个提示，你可能会得到一些稍微不同的东西。

47:04
Right? And then those models change. It's not that software doesn't change, right? To change all the time. But the documentation is really important because you want to be able to.
对吧?然后这些模型就改变了。并不是说软件不会改变，对吧？一直在改变。但是文档真的很重要因为你想要能够。

47:16
Replicate our results right? We want to be able to be sure and understand exactly how the results came to be, right when you're counting the number of times of what appears that's very strict.

复制我们的结果，对吗？我们希望能够确定并准确地理解结果是如何产生的，当你计算出现的次数时，这是非常严格的。

47:27
You can tell exactly how it is when you're trying to pick out the co locations, like the most common co locators, whether in Singapore is hot.
当你试着选择共同地点的时候，你可以准确地知道它是怎样的，比如最常见的共同地点，新加坡是否热。

47:35
Right? That's actually a very simple thing that we can do using tools every.
对吧?这其实是一件非常简单的事情我们可以用工具来做。

47:39
Exists that are not generally out, and actually, for most of us, that's usually good.
通常不存在，实际上，对我们大多数人来说，这通常是好事。

47:47
Right?
对吧?

47:47
So there's a trade off you need to make. If you want to use training, we will export. That is what we're going to do.
所以你需要做一个权衡。如果你想用培训，我们会出口。这就是我们要做的。

47:53
But do something, just be aware of. Okay.
但还是要做点什么，要意识到。好吧。

48:00
Okay.
好吧。

48:00
This is where a little bit fluent pets, I'm talking about this thing called minimal computing.
这是一个有点流利的宠物，我说的是一个叫做最小计算的东西。

48:09
All right, how do you use the least amount of calculations to achieve the goal that you operate?
好吧，你如何使用最少的计算量来达到你的目标?

48:17
Because there are these costs.
因为有这些成本。

48:20
There's a subscription cost, but there's also a sustainability cost, right?
有订阅成本，但也有可持续性成本，对吧?

48:26
You probably are all aware of the data center issue, right? The fact that we using a lot of electricity to power data centers.
您可能都知道数据中心的问题，对吧？事实是我们使用大量的电力来为数据中心供电。

48:36
And water, fresh water to cool down the surface, right?
还有水，用来冷却地表的淡水，对吧?

48:42
And as economics and global geopolitics go they tend to be different in areas.
随着经济和全球地缘政治的发展，它们往往在某些领域有所不同。

48:47

Where environmental protections or where communities don't really have the same.
在环境保护方面，或者社区没有真正的保护。

48:52
My friend.
我的朋友。

49:01
EW.
电子战。

49:02
I'll group.
我组。

49:06
You have.
你已经拥有的。

49:10
Draw.
画画。

49:12
And this is a well developed, industrialized city, and we are really running up against this issue.
这是一个发达的工业化城市，我们真的遇到了这个问题。

49:23
Of electricity provision, right? The demand is high. It's really, really high, but the cost of it.
电力供应，对吧？需求很高。它真的非常非常高，但是它的成本。

49:34

So if you have a tool that doesn't need this, and you can get the same result.
如果你有一个不需要这个的工具，你可以得到同样的结果。

49:41
Okay. So tools that are lighter, that are open, they are inspectable, versus of the black boxes.
好吧。所以工具更轻，是开放的，它们是可检查的，而不是黑盒子。

49:56
I've been talking about LMS as if the only ones that exist are like cloud in China, and.
我一直在谈论LMS，好像唯一存在的就是中国的云。

50:03
There are open source ones, so you can kind of decide, right, like, open floor. Open floor is a bit of a fan. Still, people are still using it, but they are slower, of course, they are much less well tuned.
有开源的，所以你可以决定，对吧，比如，开放的地板。开放式地板有点受欢迎。尽管如此，人们仍然在使用它，但是它们变慢了，当然，它们的调优程度要差得多。

50:14
And a lot of the commercial ones, but at least you can have a look under the good.
还有很多商业的，但至少你可以看到下面的好。

50:20
See what's going on.
看看发生了什么。

50:23
Some way, right? You get a lot more connection to what.
某种程度上，对吧？你会得到更多的联系。

50:26
They are techniques lower than us, actually, but you can run them off your English.
实际上，他们的技术比我们低，但你可以把它们从你的英语中剔除。

50:31
Of course, again, it will cost.
当然，还是要付出代价。

50:34
Actually the China, A lot of China models are really good in this, like open model, like Z, AI and.
实际上中国，很多中国模式在这方面都很好，比如开放模式，比如Z，人工智能和。

50:43
The Alibaba model, they are all open with.
阿里巴巴的模式，他们都是开放的。

50:46
So there is a diversity industry.
所以这是一个多元化的行业。

50:50
Makes sense because.
说得通，因为。

50:55
People treat these models for the reasons, and sometimes it's an economic aspect.
人们对待这些模型是有原因的，有时是经济方面的原因。

51:01
Sometimes it's a market capture perspective.
有时这是一个市场捕获的角度。

51:04
Other factors, but not this.
其他因素，但不是这个。

51:08
What you get out of it.
你从中得到了什么。

51:12
How the model is published or put useful on a chip.
如何发布模型或将其应用到芯片上。

51:17
Oh.
哦。

51:17
A big light issue, so just keep your eye on what's going on.
这是一个大问题，所以请密切关注正在发生的事情。

51:25
Okay.
好吧。

51:25
All right. So quick summary, right. LM is a black box.
好吧。快速总结一下。LM是一个黑匣子。

51:32
That simulates human like that, right? It might sound very friendly, and it's designed to be very friendly, right? It's meant to keep you.
像这样模拟人类，对吧？这听起来可能很友好，而且它的设计初衷就是很友好，对吧？这是为了留住你。

51:42
There, and then conversation. It takes on the persona seems to have emotions until you realize it doesn't, because it's kind of.
在那里，然后谈话。在你意识到它没有情感之前，它看起来是有情感的，因为它有点。

51:50
The millacium of human behavior.
人类行为的巅峰时期

51:51
That's trained to egg in very popular weeks. I was trained to keep engagement high. It's trained to be helpful, it's trained to apologize.
在最受欢迎的几周，它会被训练成鸡蛋。我接受的训练是保持高参与度。它被训练成乐于助人，它被训练成道歉。

52:00
This train to soothe your ego, and this is designed.
这列火车是为了安抚你的自尊心，这是专门设计的。

52:07
You can tell an LM that you will be happier if they were much more severe with you.
你可以告诉LM，如果他们对你更严厉，你会更开心。

52:13
Then you do it.
然后你去做。

52:15
And then sometimes that's a great way of doing it. All right, because AI can be sycopendic if you.
有时候这是一种很好的方法。因为人工智能可以是综合的，如果你。

52:22

If you sometimes I'm moving there, I'm looking at what is telling me, and I say, you're just telling me what I want to hear, right?
如果你有时我搬到那里，我看着它告诉我什么，我说，你只是告诉我我想听的，对吗?

52:29
See, and then they agree with me. So, so you have to be quite careful.
然后他们同意我的观点。所以，你必须非常小心。

52:37
There are quick questions philosophically, around the moral worthiness of an AI margin. Does it deserve moral consideration?
围绕人工智能边际的道德价值，有一些简单的哲学问题。它值得道德上的考虑吗?

52:46
Like a child, and you mistreat an AI model. Does it even make sense? Right? Life questions again.
就像一个孩子，你虐待一个人工智能模型。这有意义吗？对吧?生活又开始发问了。

52:55
I think the larger moral question is really.
我认为更大的道德问题是。

53:00
Whether it's worth the treat off.
是否值得把它拿走。

53:03
But whether this is or interested in this, and really the responsibility for verification and applying the wrong truth.
但无论这是或对此感兴趣，以及验证和应用错误真理的责任。

53:14

Providing the right context and making sure you're really double checking what the AI is giving you.
提供正确的背景，并确保你真正仔细检查AI提供给你的内容。

53:22
Um, air comes in the point of time in our society, where we.
空气出现在我们社会的某个时间点，我们。

53:29
We trust things that are easy, and we trust things that sound very proficient.
我们相信简单的东西，我们相信听起来非常熟练的东西。

53:36
So when people are eloquent, when things look eloquent and well, argue the forms of supersy, sometimes the content.
所以当人们雄辩的时候，当事情看起来雄辩的时候，争论修辞的形式，有时是内容。

53:43
And AI, in some weeks, is enhancing that, this kind of working its way through that.
而人工智能，在几周内，正在加强这一点，这是一种通过它来实现的方式。

53:51
Right? It's designed to solve the target.
对吧?它是用来解决目标的。

53:53
As historians, assets, or actually others, people in general, really ought to be a lot more clear when it comes to what it's giving.
作为历史学家，资产，或者实际上是其他人，一般来说，当涉及到它所给予的东西时，真的应该更清楚。

54:03

And it's always Look, we know this, right? So, of course, everything you read, as it somewhere else is information coming from.
而且总是听着，我们知道的，对吧？当然，你读到的所有东西，都是来自别处的信息。

54:13
Now that you know that Lmson actually pull out the source, necessarily, and beat that.
现在你知道了，Lmson实际上拿出了消息来源，并打败了它。

54:17
You really ought.
你真的应该这么做。

54:21
You can direct it in group.
你可以在小组里指挥。

54:24
Whistle.
吹口哨。

54:27
Okay.
好吧。

54:27
All right, um.
好吧，嗯。

54:29
Although bigger than pets, right in the working people talking about AI is a tool to help you explore things, or as a method to help you find.
虽然比宠物大，但在人们谈论人工智能时，它是一种帮助你探索事物的工具，或者是一种帮助你找到东西的方法。

54:40
And so I just want foreground that clarity is very important for us.
我想说的是清晰度对我们来说很重要。

54:46
People who use AI in research, right? Are you using it to ask the fundamental basis of your method? Is it?
在研究中使用人工智能的人，对吧？你是在用它来询问你的方法的基本基础吗？是吗？

54:55
Key part of your method, or is it really a helping tool?
你的方法的关键部分，或者它真的是一个辅助工具吗？

55:00
For you to do some of the work that.
让你做一些工作。

55:02
You explosives.
你炸药。

55:04
It's much more secure on the two sides. It's much less secure than that. Just.
两边都安全多了。它远没有那么安全。而已。

55:11
Pretty handsome.
很英俊。

55:12
Alright, we'll have a look, a couple of bus works that you will come across.
好了，我们来看一下，你们会遇到的一些公共汽车作品。

55:20
You look at you. Supervised was a supervised.
你看看你自己。被监督就是被监督。

55:25
Right? Supervisor, is what we're talking about in terms of giving it sort of ground truth and telling it the facts of the matter, right? This is right. This is ground. This is cat as a dog, right?
对吧?监事，就是我们所说的给它一些基本的事实告诉它事情的真相，对吧?没错。这是地面。这就像猫和狗一样，对吧?

55:36
That's training. We call it training, and that's super isolated, right? So we?
这是培训。我们称之为训练，这是非常孤立的，对吧？所以我们吗?

55:42
L envys us an answer. We read it and keep it back. We went one way.
我很羡慕我们的答案。我们阅读它并保留它。我们走了一条路。

55:49
Right? And then it recalibrates and it tries again. So this is supervised training, right? It's like having a teaching reading. You are the teacher making.
对吧?然后重新校准，再试一次。所以这是有监督的训练，对吧？这就像有一个教学阅读。你是老师的成就。

55:58
The students is recorded as wrong. And then there's unsupervised which is you just send it off there in the city.
这个学生被记录为错误。还有一种是无监督的，就是你把它送到城市里。

56:05
Tell me what there is.
告诉我有什么。

56:06
Kind of good advice. So you can see that a little bit with topic modeling. You remember topic modeling, the box of rebellion one, right?
不错的建议。你可以从主题建模中看到一点。你还记得主题建模，叛逆的盒子，对吧?

56:19
Who run the topic modern two, and it brings up nine different groups and topics, and things are the most dominant.
谁经营的主题现代二，它带来了九个不同的群体和主题，和东西是最主要的。

56:29
Conceptual reps that exist concepts, groups of concepts that exist in the people.
存在的概念代表，存在于人体内的概念群。

56:35
That's unsupervised.
这是无监督。

56:36
You're not telling what to do, drawing the.
你没有告诉我该怎么做，画。

56:40
On explore.
在探索。

56:42
Every week.
每一个星期。

56:48

Okay.
好吧。

56:48
And they pause here, how's everyone do everything?
他们停在这里，大家是怎么做的?

56:54
Okay. Agenda. AI. This is egenic AI. Egenic AI is the bus that's been around. Probably.
好吧。议程。人工智能。这是基因人工智能。基因人工智能是一直存在的公共汽车。可能。

57:06
So does anyone know what I think? Does anyone use?
有人知道我是怎么想的吗？有人使用吗?

57:10
Which is so. So we've been talking about Gent ei, right? Which is a chat problem, right? Open up a chat window.
事实的确如此。我们一直在讨论genei，对吧？这是一个聊天问题，对吧？打开一个聊天窗口。

57:22
Having a problem, can get a response, right, then look like a problem in the conversation. Okay. Ajectic AI is kind of like if you have an intern working with you.
有问题，可以得到回应，对吧，然后在谈话中看起来像个问题。好吧。ajtic AI就像你身边有个实习生。

57:32
And you tell it, okay, in turn, you have access to this set of documents and material. Your role is to summarize them for me.
然后你告诉它，好吧，反过来，你可以接触到这些文件和材料。你的任务是给我总结一下。

57:42
Right? Every time I put in a new document, summarize it for me, or transform it, or turn it into a PDF, something like that, you have a very specific role.
对吧?每次我放一个新文件，给我总结一下，或者转换一下，或者把它变成PDF文件，诸如此类，你都有一个非常具体的角色。

57:50
All right, why gyros is very dangerous? It's because people use it for cooling on live databases on actual projects.
好吧，为什么陀螺很危险？这是因为人们用它来冷却实际项目中的实时数据库。

57:59
You say, oh, jane, ai, please help me redo this code so that it does this. And then Jen, air goes, oh, okay. The easiest way to solve this problem is to delete everything.
你说，哦，jane, ai，请帮我重做这段代码，让它这样做。然后珍，空气变了，哦，好吧。解决这个问题最简单的方法就是删除所有内容。

58:09
It's everything, and it's you recover. It deletes the backup as well. It doesn't really solve the problem, right? So it has happened.
这是一切，这是你的恢复。它也会删除备份。这并不能真正解决问题，对吧？所以它发生了。

58:19
Many times documented where people have lost the link, E.G., agentic, AI, to their emails.
很多时候，人们失去了链接，例如，代理，人工智能，到他们的电子邮件。

58:27
Summary. Work this clear out the span. Do this, do that, you know, follow certain rules. I don't want to be so busy, but they did everything.

总结。算出张成的空间。做这个，做那个，遵守一定的规则。我不想这么忙，但他们什么都做了。

58:36
So.
所以。

58:40
Very risky, right? But you must work, because agents don't work alone.
很冒险，对吧？但你必须合作，因为特工们不会单打独斗。

58:48
Have a group of agents. One agent does the research work, one agent does the organization work, one agent does the summary work, one agent does whatever. Then you say, agent A is doing this. Agent B is doing that. Agent B and A, you ought to work Together.
有一组代理人。一个代理做研究工作，一个代理做组织工作，一个代理做总结工作，一个代理做任何事情。然后你说，探员A在做这个。Agent B就是这么做的。探员B和探员A，你们应该合作。

59:02
To achieve this goal. So they are not semi autonomous. They're kind of working autonomously without your own direct supervision.
实现这一目标。所以它们不是半自主的。它们可以在没有你直接监督的情况下自主工作。

59:12
And when you connect them to your documents, your drive, you know, all your material and all that, your code, it can be very powerful.
当你把它们连接到你的文件，你的驱动器，你所有的材料和所有的东西，你的代码，它会非常强大。

59:19
Because now you don't have to quote to do any of that.
因为现在你不需要引用这些了。

59:23
I want you to do this.
我要你这么做。

59:27
And then we will do it.
然后我们来做。

59:28
We would like magic.
我们喜欢魔术。

59:30
Right?
对吧?

59:30
Also waiting, because you just delete everything.
还在等待，因为你刚刚删除了一切。

59:37
That's it, okay. But there's the difference between gen, AI, AI.
就是这样。但是gen， AI， AI之间是有区别的。

59:45
Because it has access to more things.
因为它能接触到更多的东西。

59:48
Ultimate multiple processes for you. You give it a role.
为您提供终极多重流程。你给它一个角色。

59:53
That's why some teaching. Okay.

这就是为什么有些教学。好吧。

01:00:01
Questions.
的问题。

01:00:01
Big password.
大的密码。

01:00:05
Lots of training courses, a lot you think in the air to save a life.
很多培训课程，很多你认为在空中拯救生命。

01:00:12
But of course, these are on the spectrum, right?
当然，这些都是光谱上的，对吧?

01:00:17
These are typologies. I will show you actually how we kind of do it.
这些都是类型学。我会向你们展示我们是怎么做的。

01:00:22
Use.
使用。

01:00:23
You've all got the announcement to link. Correct. There are two bots in there on different consequences materials. Those are actually the AI, but wrong.
你们都有公告要链接。正确的。有两个机器人在不同的结果材料上。这些其实是人工智能，但错了。

01:00:33
To be a bit more authentic.

更真实一点。

01:00:34
Okay, we'll show you.
好的，我们会展示给你看。

01:00:37
Okay.
好吧。

01:00:39
So here are the thoughts. If you've not had a look, we'll give everyone a little bit of time.
下面是我的想法。如果你还没看，我们会给每个人一点时间。

01:00:44
To explore. I use the one on the left here. This is the correspondence of.
去探索。我用左边的这个。这是的对应关系。

01:00:49
Also of excess, maybe, except Dylan, because you don't have any ID, but, or use this control.
也可能是过量的，除了迪伦，因为你没有任何ID，但是，或者使用这个控件。

01:00:56
Marriage anyway.
婚姻无论如何。

01:01:08
Have fun.
玩得开心。

01:01:14
About 5 min where you do free around, put in the problem, see what

happens, the one that left. But if you want, you can try to want to write. Listen.
大约5分钟的时间，你可以自由走动，把问题放进去，看看会发生什么，剩下的那个。但是如果你想的话，你可以试着去写。听。

01:01:23
So that's for the novels, and this is for physical response.
这是小说，这是身体反应。

01:02:34
Was it?
是吗?

01:02:36
Oh.
哦。

01:02:36
So we're getting our response.
我们得到了响应。

01:02:44
Yes, I want to log in.
是的，我想登录。

01:02:48
Are you getting Okay? So it is working.
你还好吗？所以这是有效的。

01:02:59
For about 5 min, basically explore a little bit.
大概花5分钟，稍微探索一下。

01:03:32
To the other boy.

给另一个男孩。

01:03:33
That's all right. I also.
所以你做了什么

01:04:05
Topic.

01:04:06
1846, two, 69. And there's a third.
1846, 2, 69。还有第三个。

01:04:09
Volume of overlapping and supplemental letters, which is additional.
重叠和补充字母的数量，这是额外的。

01:04:13
Correspondence. Okay.
信件。好吧。

01:04:16
So that's the data set.
这就是数据集。

01:04:17
So that's the coppers that is going on. We know, of course, LMS have its own main coppers, right? But you can see that it has access to itself three volumes.
这就是铜的作用。我们当然知道LMS有它自己的主铜，对吧？但是你可以看到它有自己的三卷。

01:04:28
Of letters and correspondence. Okay.
信件和通信。好吧。

01:04:33
That's what I supplied. That is the material, the extra purpose that the LM didn't have necessarily access to.
这是我提供的。这是材料，是登月舱没有必要用到的额外用途。

01:04:46
Right? So privilege, specific information that doesn't affect us. So what I did was I gave it the three volumes of correspondence.
对吧?所以特权，不影响我们的特定信息。所以我给了它三卷的信件。

01:04:56
So that's the DSA question. The second question is, how does the AI know how this one?
这就是DSA的问题。第二个问题是，人工智能是怎么知道这个的?

01:05:00
Because you fully operate the ID form, right? And your responses don't always sound like this.
因为你完全操作了ID表单，对吧？你的回答并不总是这样的。

01:05:06
I told it exactly how to respond to the question.
我准确地告诉了它如何回答这个问题。

01:05:23
When a user uses this box, how should it be responded?
当用户使用此框时，应该如何响应?

01:05:28
Answers here. And the essays are, there are ten, two text novels that I took from for the written book, and then the three volumes of its correspondents, which are also like Your.
答案在这里。这些散文，有十篇，两篇是我从书面语小说中摘录的，然后是它的三卷通讯，也像你的。

01:05:40
Right? A little bit, actually, very good, soft.
对吧?一点点，非常好，很软。

01:05:42
And this is my problem.
这就是我的问题。

01:05:48
There's a cool problem. Well, actually, it's not a cool problem because there's actually a lot more, right? But maybe.
有一个很酷的问题。实际上，这不是一个很酷的问题因为实际上还有很多，对吧？但也许。

01:05:55
Worth reading it out a little bit. Okay. So what I've done is I've given a role to.
值得读一下。好吧。所以我所做的就是给。

01:06:02
The AI. You are a reader with access to the letters of tons about which that PET files are your source material than I described.
人工智能。你是一个读者，可以接触到大量的信件，而PET文件是你的原始材料，而不是我描述的。

01:06:17
It keeps asking just because I kept miswriting. Charles.
它一直在问，因为我一直写错。查尔斯。

01:06:23
Because it's tongue down with And so I think it gets to the confused. But.
因为它的舌头向下所以我觉得它会让人感到困惑。但是。

01:06:28

I was looking at another dialing thing. I liked a lot of dialing. And then, without writing this draw.
我在看另一个拨号程序。我喜欢不停地拨号。然后，不用写这个图。

01:06:42
Okay, so you can see where they came out.
好了，你可以看到它们是从哪里出来的。

01:06:44
Something. So just replace down the size of the difference. You are a leader, right? So I'm telling the LM to perform the mark.
某物。所以只要把差的大小替换下来。你是一个领导者，对吗？所以我告诉LM执行标记。

01:06:51
The one that I've given to. LM Okay.
我给的那个人。LM好。

01:06:54
You know, with access to the local Charles Second.
能接触到当地的查尔斯二世。

01:06:59
The attached files are your source material. The treatise contain letters from the author, charles Chicken Volume. Source Tree.
附件是你的原始资料。这篇论文包含了作者查尔斯·肯奇的书信。源树。

01:07:05
Review the information for general teams to such as to the counter two.
检查一般小组的信息，如到2号柜台。

01:07:09
Not straight, too far from the context provided by the Compass. So I put it specifically to tailor with responses based on the attached files, the documents I've given. It.

不直接，离罗盘提供的上下文太远了。所以我把它特别定制与响应基于附件的文件，我给的文件。它。

01:07:19
Right? Um.
对吧?嗯。

01:07:23
Instruction one respond is the reader of Dickens. Providing information about the content is requested, but only based on material provided content source.
指示一回应是狄更斯的读者。要求提供有关内容的信息，但仅基于所提供的内容源的材料。

01:07:30
Indicate where you don't have the right information, right? So.
指出你没有正确信息的地方，对吧？所以。

01:07:33
Think it way. It just isn't. You don't have access maintain a tone that is knowledgeable and informative.
这样想。事实并非如此。你没有机会保持一种知识渊博和信息丰富的语气。

01:07:40
Does the artful foundations can respond convolationally? But first, the conduct politely, who was asking for the committee.
巧妙的基础是否可以做出卷积性的反应？但首先，行为礼貌，谁是委员会的要求。

01:07:47
So you're really constraining the behavior of the AI in terms of where trying to receive information from how it ought to be speaking, how it ought to work with different.
所以你真的在限制人工智能的行为，从哪里接收信息，它应该如何说话，它应该如何与不同的人一起工作。

01:07:59
It's right. So uses the.
它是对的。所以用了。

01:08:01
That don't necessarily engage with the material. Student back.
不一定和材料有联系。学生回来。

01:08:04
Okay. Construction. Two retrieve extensive, longer quotations from the corpus that are really adjusted periods, right? You will notice that it's given you complete.
好吧。建设。二是从语料库中检索大量的，较长的引文这些引文都是经过调整的时期，对吧？你会注意到它是完整的。

01:08:16
Passages rather than brief excerpts, right? Because for us, as the science context is really important, how work is papers.
段落而不是简短的摘录，对吧？因为对我们来说，科学背景非常重要，如何工作就是论文。

01:08:22
Within the paragraph, or the pieces of its kind of important spot.
在段落内，或各片段中有它那种重要的点。

01:08:28
Understand.
理解。

01:08:28
With us. AI.
和我们在一起。人工智能。

01:08:30

Um, keep contact with text around Keeper things concept, so if it says robbery, right or.
保持与守护者相关的信息联系，如果上面写着抢劫，对吧。

01:08:42
The word.
这个词。

01:08:44
A salt, right was there as the context to that curve.
盐，就在那里作为曲线的背景。

01:08:52
So.
所以。

01:08:52
Be helpful, right?
要乐于助人，对吧?

01:08:54
Based on the topics, we treat multiple relevant quotations on the topic from different options. So.
基于主题，我们从不同的选项中处理该主题的多个相关报价。所以。

01:09:01
Don't just unless it isn't.
除非不是，否则不要这样做。

01:09:04
Perfect logifications for the shot the loss.
完美的逻辑为射击失败。

01:09:08
So.

所以。

01:09:09
Because they give us a bit more simple part.
因为题目给出了更简单的部分。

01:09:13
And length is a good proxy for, sort of, how much the authors have you going into the concept.
长度是一个很好的代表，某种程度上，作者让你对这个概念了解了多少。

01:09:22
Working with them. So this is the adjudication that I need to do in historian is like, what are the most useful things that I'll get out from my office? Longer quotations, contextualized quotations.
和他们一起工作。所以这就是我在做历史学家时需要做的判断，比如，我从办公室里得到的最有用的东西是什么？更长的引文，语境化的引文。

01:09:34
Quotations that come directly from thoughts that also look from across different parts of the conference, like one, just one bit of the focus.
引用直接来自会议不同部分的思想，就像一个，只是一个焦点。

01:09:43
Okay.
好吧。

01:09:43
And then there's way more, actually, I think this is only 25 % of the problem. There's another 75 % of the how the quotational comment is, how you structure the explanation.
还有更多，实际上，我认为这只是问题的25%。还有75%是关于引用评论的，你是如何组织解释的。

01:09:54

And how you do the summary at the end.
以及你如何在最后做总结。

01:09:56
Having a focus on academic decisions.
专注于学术决策。

01:10:00
Um.
嗯。

01:10:01
Instruction to acknowledge when there's not enough information on compass.
当罗盘上没有足够的信息时，指示要承认。

01:10:05
So these are, this is a very, very long and detailed prompt that actually sits behind what you're seeing.
这是一个很长很详细的提示实际上就在你所看到的后面。

01:10:13
What has happened, actually, is this is a custom box, right? It's got a bigger spot away, right?
实际上，这是一个定制的盒子，对吧？它有一个更大的位置，对吧？

01:10:21
What has happened is, I have taken an off the shelf in this his clock open six, right? I had given it this promptus and the documents.
事情是这样的，我从架子上拿了一个他的钟开了6点，对吧？我给了它这个提示词和文件。

01:10:31
And then I say, okay, start the conversation here after this prompt. That's where you guys come in. When you log in, this promise in the conversation.

然后我说，好吧，在这个提示之后开始对话。这就是你们的作用。当你登录时，这个承诺在对话中。

01:10:40
The.
这个。

01:10:42
Matures in the conclusion you're just not seeing enough. It comes just before I asked it, how do you know how to respond? And it's given me, like, different guidelines, which is matching to what you can.
在结论中成熟，你只是看得不够。它就在我问它之前出现，你怎么知道该怎么回答？它给了我不同的指导方针，这是与你所能匹配的。

01:10:55
Destroying directly from the instructions I've given up.
根据我给出的指令直接销毁。

01:10:58
You can actually, you can ask the I why respond to the possibility of the shelf? AI.
实际上，你可以问为什么对架子的可能性做出反应？人工智能。

01:11:06
They would respond to.
他们会回应。

01:11:08
You give you a sense, I have actually got into the English version.
你给你一个感觉，我实际上已经进入了英语版本。

01:11:13
Because the, what in your study for staff is they have.
因为在你对员工的研究中，他们有。

01:11:19
The office of versions of C clock of Gemini.
双子座C时钟版本的办公室。

01:11:27
If added such instructions, like, what did I discover? Refer to n US documents where possible.
如果加上这样的说明，比如，我发现了什么？在可能的情况下，请参考美国文档。

01:11:35
Summarize the last five compositions that this user has had in users as contexts.
将该用户在用户中拥有的最后五个组合作为上下文进行总结。

01:11:42
Right? And so when I see things popping up in my conversation and like, I'm not quite sure where they got this information from. So I does a bit deeper, because it started giving me an example that I talked about five.
对吧?所以当我看到在我的谈话中突然出现的东西，我不太确定他们从哪里得到这些信息。所以我做得更深一点，因为它给了我一个我讲过的例子。

01:11:54
Conversation is a goal now thinking, why is this coming up? That means there must be some context that was added in. So I started inquiring. I was like, what are your Instructions.
对话是一个目标，现在想想，为什么会出现这个？这意味着必须添加一些上下文。于是我开始询问。我说，你的指导是什么。

01:12:03
Right. Why did you pick this specific example? And is it, I have to be honest with you.
正确的。为什么选择这个特定的例子？是吗，我得跟你说实话。

01:12:09

Very.
非常。

01:12:13
An underlying level of instructions that was customized. And so the same way that you guys don't see this instruction there was.
自定义的底层指令。同样的道理，你们没有看到这里的说明。

01:12:20
Actually another self instruction that anyone had encountered before the window came to them. Wow. And you can define that. You figure that out somehow when you look in.
实际上是另一个自我指导，每个人都遇到过在窗口来到他们面前。哇。你可以定义它。当你往里看的时候，你就知道了。

01:12:32
And you see, like, why is this giving increase? The instructions have left an imprint on the condition.
你会发现，为什么捐赠增加了？这些指示给情况留下了深刻的印象。

01:12:38
Of course, there are a lot of gut rules around things like.
当然，有很多直觉规则，比如。

01:12:42
Mental health susceptor prevention when it comes to really sensitive topics, the LMS themselves, the companies have got rails, but in us, and stuff has always.
心理健康易感者的预防当涉及到非常敏感的话题时，LMS本身，这些公司都有问题，但在我们身上，事情总是如此。

01:12:54
Instructions, which is quite interesting.
说明，这很有趣。

01:12:59
We're here to get up right. Of course, a lot of times they're not going to tell you, in democracy, you provide the information.
我们来这里是为了振作起来。当然，很多时候他们不会告诉你，在民主制度下，你提供信息。

01:13:07
Like that's specifically cool enough.
这就足够酷了。

01:13:09
You can actually have a sense, at least for some of these underlying infrastructure and architecture of the models.
您实际上可以有一个感觉，至少对于这些底层的基础设施和模型的体系结构。

01:13:19
We're playing around it.
我们只是玩玩而已。

01:13:23
Okay, I'm not here, right? We just talked a little bit about this, and we've covered some of this.
好吧，我不在这里，对吧？我们只讲了一点，我们已经涵盖了一些。

01:13:32
But your data set essentially needs to be a form that the LMS can understand. With text is very easy because they're trained on text, right? And so working with text analysis, corporate analysis is.
但是你的数据集本质上需要是LMS能够理解的形式。使用文本很简单，因为他们受过文本训练，对吧？所以使用文本分析，企业分析。

01:13:44
A little bit, triviola to them.
一点，对他们来说是微不足道的。

01:13:46
They can query the three volumes of letters that would have taken you maybe a month to really get through, and they immediately pull out starters relevant for you.
他们可以查询你可能要花一个月才能看完的三卷信，他们会立即拿出与你相关的开头。

01:13:54
Right? Not the similar to some of the text answers that we've been using.
对吧?和我们一直使用的文本答案不一样。

01:14:00
Counting work, you can see, right, looking at co location, identifying particular terms, right? The stuff that we did in seminar form.
计算工作，你可以看到，对吧，关注co - location，识别特定的术语，对吧?我们在研讨会上做的事情。

01:14:07
But can we rely on the summaries that it does?
但我们能相信它的总结吗?

01:14:12
Good question depends on how low bearing you want it to be.
好问题取决于你想要多低的轴承。

01:14:17
So the way that this book was written is such that you have the original locations of where these come from, so they help you do the distant reading.
所以这本书的写作方式是这样的，你有这些东西的原始出处，所以它们帮助你进行远读。

01:14:27
The close reading, you stood for yourself.

细读，你坚持自己的立场。

01:14:29
So there's a tool, there's a tool.
有一个工具，有一个工具。

01:14:33
To help you explore the cultures. It's not infallible, but there's a very easy way to do it.
帮助你探索不同的文化。这不是绝对正确的，但有一个很简单的方法。

01:14:41
Without having to learn poems, E.G.?
不用学诗，例如?

01:14:43
Some audiences, you can give it right. If you're using it to publish a paper, then it becomes a lot more difficult.
有些观众，你可以给它正确的。如果你用它来发表一篇论文，那就变得困难得多了。

01:14:59
Right? This is difficult to replicate, necessarily.
对吧?这必然是难以复制的。

01:15:05
But you're trying your best to softly, not commit yourself against the dangers of L and hallucination.
但你在尽自己最大的努力，不让自己陷入L和幻觉的危险之中。

01:15:11
You wrong. Information drawing from the context of Singapore, right?
你错了。信息来源于新加坡的背景，对吧?

01:15:16

So it's good for exploring things they are, in some ways, good and helpful. Users of gnai for.
所以这对探索事物是有益的它们在某些方面是有益的。gnai的用户。

01:15:25
Things like quantitative analysis. They can also do quantitative analysis, but it's not a bad they're quite decent at it. They can run statistical tests because they're just calculating numbers.
比如定量分析。他们也可以做定量分析，但还不错，他们做得很好。他们可以进行统计测试，因为他们只是在计算数字。

01:15:36
Great, and those are things that computers are really good at. There's no judgment call.
很好，这些都是电脑擅长的。没有评判的余地。

01:15:40
So when you depend on a computer to do judgment, course like summarizing, then that's where.
所以当你依靠电脑来做判断时，当然，比如总结，那就是。

01:15:50
Of course. I mean, at the very base of your data set needs to be legible, like to the computer. He needs to be computer beautiful.
当然可以。我的意思是，在你的数据集的基础上，需要像电脑一样清晰可辨。他需要像电脑一样漂亮。

01:16:03
Right? You can give it an image of untranscribed text, like a scanner for a document, and expect it to.
对吧?您可以给它一个未转录文本的图像，就像一个文档扫描仪，并期望它。

01:16:13
Don't be able to read it right?
看不懂，对吗?

01:16:14
They might have some clue. You start doing the job. You cannot do it very close and so forth. But the cleaner, remember, garbage in garbage job, right?
他们可能有一些线索。你开始工作。你不能做得很近等等。但是清洁工，记住，垃圾在垃圾工作，对吧?

01:16:23
Cleaner the stuff that you put in your data, the better your design, more likely to do that.
你在数据中放入的东西越干净，你的设计就越好，这样做的可能性就越大。

01:16:28
So again, same principles as digital history.
所以，和数字历史的原理一样。

01:16:39
General.
将军。

01:16:39
Um, tolerance may not be very high. It can accommodate some errors, so less can be very useful if you have Osalus, slightly dirty.
你的容忍度可能不是很高。它可以容纳一些错误，所以如果您有Osalus，那么less非常有用，稍微有点脏。

01:16:52
But then you also be quite very because if it is, it dirty, because the scanning is different.
但你也会非常，因为如果它是，它很脏，因为扫描是不同的。

01:16:58
And LM Patrina, modern language, modern English.
还有LM Patrina，现代语言，现代英语。

01:17:02
Are you selling the correcting things that like?
你是在卖你喜欢的矫正用品吗?

01:17:06
But you actually your research question.
但实际上你的研究问题。

01:17:11
This is about language deviation, and then LM is about tending to experience media.
这是关于语言偏差，然后LM是关于倾向于体验媒体。

01:17:17
Of language, and that very it goes comfort to what you're looking for.
语言，这是非常舒适的，你正在寻找。

01:17:22
Where it's blackening the language.
让语言变黑。

01:17:25
And you don't want that. You actually want to document the variability of that.
你不希望那样。你需要记录下它的可变性。

01:17:29
Don't know good to but you have to make that decision, right? You need to understand how the L and X, you need to understand what your research method is. And.
不知道该怎么做，但你必须做出决定，对吧？你需要了解L和X，你需要了解你的研究方法是什么。和。

01:17:39

And whether this term care, government or not.
不管这个词是否属于政府。

01:17:41
Okay, um.
好的,嗯。

01:17:47
Different models have different personality, different quotes. So if you clear on, then get a little bit of the sense of what they tend to do.
不同的模型有不同的个性，不同的报价。所以如果你明白了，就能对他们倾向于做什么有一点了解。

01:18:00
You can choose a model that is a bit closer to what you are thinking about.
你可以选择一个更接近你所想的模型。

01:18:07
I know humanities people prefer plots just because it's a bit more interrogated. It doesn't try to quickly summarize things for you and make it peanuts.
我知道人文学科的人更喜欢情节，因为它更有审问性。它不会试图快速地为你总结事情并使它变得微不足道。

01:18:17
Like GP or even China.
比如GP，甚至中国。

01:18:19
That personalities, and not really clear with deepsit, but if anyone has experience with Deeps.
我不太清楚deepsit的个性，但如果有人用过deepsit。

01:18:26
Um.

嗯。

01:18:28
All right, um, nothing. Break.
好吧，没什么。休息。

01:18:37
Come back.

01:18:45
If you look at the.
如果你看一下。

01:18:47
As pets, you will realize that any work that has to do with AI is giving you a lot.
作为宠物，你会意识到任何与人工智能有关的工作都会给你带来很多好处。

01:18:53
All within the last few.
都在最后几分钟。

01:18:56
And there's still a lot of thinking and your writing.
还有很多想法和你的作品。

01:18:58
Going on around our area.
发生在我们这一带。

01:19:01
Which gen AI fits within our view and our cultures. It does not run content in many ways, but then they're also kind of productive.
哪一代人工智能符合我们的观点和文化。它在很多方面都不运行内容，但它们也很有成效。

01:19:11
Using Jenny, I in very specific cases. But all the bigger and pets talk about source criticism, like what's in the culture from what's missing. And there's something that we've been talking about since the.
在非常具体的情况下使用Jenny。但所有的大公司和宠物都在谈论源批评，比如文化中有什么，文化中缺少什么。有件事我们一直在讨论，自从。

01:19:22
Right? What's in the data set, is it to be representative?
对吧?数据集中有什么，它有代表性吗?

01:19:28
Of the reality of trying to describe, right?
试图描述的现实，对吧?

01:19:31
Data criticism, which is kind of the new layer that you may not have thought about when, until you started digital history, right? How is the data digitized? How is it structured? How is it clean?
数据批评，这是一种新的层面，你可能没有想过，直到你开始研究数字历史，对吧？数据是如何数字化的？它的结构是怎样的？它是如何清洁的?

01:19:43
Right is very specific to the format of the sources that we use, in particular, which has.
正确是非常具体的格式，我们使用的来源，特别是有。

01:19:50
To be digitized in some form, right? And opposed to digitization, has all sorts of different structure factors that.
以某种形式被数字化，对吧？而反对数字化的，则有各种不同的结构因素。

01:19:57
The way to understand the ways understood.

理解的方式理解的方式。

01:20:01
What we are kind of walking towards in the two section where we started with the text analysis from of JS.
我们从JS的文本分析开始，在这两个部分中我们正在走向什么。

01:20:11
By thinking about the criticism around the tool itself.
通过思考围绕工具本身的批评。

01:20:14
What does the model do in this case for AI? What does the model do?
在这种情况下，模型对人工智能有什么作用？这个模型是做什么的?

01:20:17
What is and.
什么是和。

01:20:18
You see, and understand what he's doing right?
你明白他在做什么，对吧?

01:20:22
With text analysis, to the open source stuff, and go to the lab. That was very obvious, know exactly how it's done. Which end it becomes a lot more efficient.
用文本分析，去开源的东西，然后去实验室。这很明显，我知道该怎么做。这一端的效率会更高。

01:20:31
Right? And so then that might change your decision about how and where they might want to use it.
对吧?因此，这可能会改变你的决定，关于他们如何以及在哪里使用它。

01:20:38
Can we trust the tool that we are using to read?
我们能相信我们用来阅读的工具吗?

01:20:42
So keep that. We've given it.
所以留着吧。我们已经给出了。

01:20:44
All right, how reliable you think the truth is?
好吧，你觉得真相有多可靠?

01:20:47
Doing the job that you want to compete? Is it an appropriate?
做你想要竞争的工作？这合适吗?

01:20:51
And in that case, in that sense, it's not a new question, right? There are some questions that, you know, cannot be asked by a particular.
在这种情况下，从这个意义上说，这不是一个新问题，对吧？有些问题，你知道，不能由一个特定的人来问。

01:20:58
You just use another two, right? Or you might use another two.
你只用另外两个，对吧？或者你也可以用另外两个。

01:21:02
But this question of trust is an interesting one, because how is it used in an institution, in an?
但是信任是一个有趣的问题，因为它是如何在一个机构中使用的?

01:21:12
Organization? How is it designed or sort of, big factors in terms of whether two is adopted or not?
组织?它是如何设计的，或者是是否采用两个的重要因素?

01:21:18
May end up working in organizations that are very poor AI might end up working in organisms that are suspicious of AI, right?
可能最终会在人工智能很差的组织中工作可能最终会在怀疑人工智能的组织中工作，对吧?

01:21:27
Um, it will vary a lot, at least very much.
嗯，会有很大的不同，至少是很大的不同。

01:21:31
So getting to actually understand what AI is, then AI is, and then figure out to yourself, actually, when it's meeting up.
所以要真正理解AI是什么，然后AI是什么，然后自己弄清楚，当它遇到的时候。

01:21:39
Good idea. Oh, well, maybe it can be very helpful.
好主意。哦，好吧，也许它会很有帮助。

01:21:43
But use, the use of AI is a social process. In many ways, your attributes to AI are shaped by the people who are around you.
但是人工智能的使用是一个社会过程。在很多方面，你对人工智能的看法是由你周围的人塑造的。

01:21:53
And your own experience with it. And so Joan and Rees are talking about this in the reading that you read.
以及你自己的经历。所以Joan和Rees在你们读的阅读材料中谈到了这一点。

01:21:58
About AI use in humanities.
关于人工智能在人文学科中的应用。

01:22:04
Something that came up in the paragraph foundation is obviously.
在段落基础中出现的一些东西显然是。

01:22:10
On the social perception of Gentlemen.
论绅士的社会观念。

01:22:13
I.
我。

01:22:14
People looking for human authenticity, rather than talking to.
人们寻找人类的真实性，而不是交谈。

01:22:20
In the university, something really interesting, especially when it comes to student reading. Students absolutely hated when teachers use air to create papers.
在大学里，一些非常有趣的事情，特别是当涉及到学生阅读。学生们非常讨厌老师用空气做试卷。

01:22:31
And teachers absolutely hate it when they're reading stuff that seems pretty obvious that students use ya to.
老师们非常讨厌他们读的东西，这些东西很明显是学生们用来读的。

01:22:40
To do. And I think a lot of it, I mean, a lot of it is intellectual honesty, right? But a lot of it also has to do with a relationship that just has.
要做。我认为很多，我的意思是，很多都是智力上的诚实，对吧？但这也与一段关系有很大关系。

01:22:50
Guest students. In terms of the.
客人的学生。用。

01:22:54
Own emotional investment in that relationship, that we are reading, pieces of work that do come from students and what they think.
我们在这段关系中的情感投入，我们正在阅读的，来自学生的作品和他们的想法。

01:23:03
It reflects because.
它反映了原因。

01:23:04
What is a proxy for understanding how students are thinking? That's what I'm saying. Example.
什么是理解学生思维方式的代表？这就是我要说的。的例子。

01:23:08
We're not really here to gauge. We fit each understanding, but assignment for us teach that.
我们不是来衡量的。我们符合每一种理解，但作业教我们这些。

01:23:16
Right? The same way that students want to know what teachers actually thinking, it goes off.
对吧?就像学生想知道老师到底在想什么一样，它会熄灭。

01:23:22
Feedback in grades, right? You don't want a machine to replace that, because then it's not.
分数反馈，对吧？你不会想要一台机器来代替它，因为那样它就不是。

01:23:27

So there's a question of authenticity. It really underpins a lot of this discussion around how we think about your answers.
所以有一个真实性的问题。它确实支持了很多关于我们如何看待你的答案的讨论。

01:23:34
And this all happens.
这一切都发生了。

01:23:37
All right.
好吧。

01:23:42
Here is a piece of Texas correspondence letter. This is why I confused it with Darwin, because this is a Darwin letter.
这是一封德州的函件。这就是为什么我把它和达尔文搞混了，因为这是一封达尔文的信。

01:23:55
Um.
嗯。

01:23:59
And what we used to do before gen AI is we will use transcription tools, handwritten text transfer recognition, HDR.
在人工智能时代之前，我们会使用转录工具，手写文本传输识别，HDR。

01:24:09
You would find it in places like transcripts. You may have come across transcripts.
你会在成绩单之类的地方找到它。你可能遇到过成绩单。

01:24:14
It's a platform with different models, different algorithms for recognizing

chemical texts.
这个平台有不同的模型，不同的算法来识别化学文本。

01:24:21
First time, I think the first across centuries, even in the same language, our conventions are different, right?
第一次，我想这是几个世纪以来的第一次，即使是在同一种语言中，我们的习惯是不同的，对吧?

01:24:27
But you can e.g., use this and smog with something like this. This machine.
但是你可以用这个和烟雾来搭配这样的东西。这台机器。

01:24:32
Obviously, there are errors in here that we.
很明显，这里有一些错误。

01:24:36
Really need to be.
真的需要。

01:24:39
Corrected. I've actually corrected this part here, but this part here is clearly still very good.
纠正。我已经修正了这部分，但这部分显然还是很好。

01:24:47
And this is so supervised training, right? If you have a collection of a hundred letters by the same person, then you might invest your time in correcting or modeling training it to read this person's letters.
这是有监督的训练，对吧？如果你收集了同一个人写的一百封信，那么你可能会花时间去纠正或训练它来阅读这个人的信。

01:25:01
You read, they are specific handwriting, right? All handwritings are

different. There is some convention that is shared across the society what we do.
你看，它们是特定的笔迹，对吧？所有的笔迹都不一样。我们所做的事在社会上是有一些共同的惯例的。

01:25:11
A hundred years or so in three years, right? And so there's a general, applicable algorithm, but to read a specific person can write from the nuance.
三年内一百年左右，对吧？所以有一个通用的，适用的算法，但要读一个特定的人可以从细微差别中写出来。

01:25:19
Weekly training, right? So we used to do this kind of training a bit more.
每周训练，对吧？所以我们过去经常做这种训练。

01:25:24
Those, um.
这些,嗯。

01:25:25
Can generate something of this process.
可以产生一些这样的过程。

01:25:31
Right? We can try and add it somewhere into this pipeline where we are cleaning the detail.
对吧?我们可以尝试将它添加到我们正在清理细节的管道中。

01:25:38
I was working a little bit on this, looking at this text, I was created by OCR, HDR, and seeing whether we can.
我在研究这个，看这个文本，我是由OCR， HDR创建的，看看我们是否可以。

01:25:46
Correct it automatically in some way.
以某种方式自动纠正它。

01:25:49
What I guess my section would be, would be to.
我想我的部分应该是。

01:25:55
Try and make everything process from the LM a lot more visible. So getting it to report. I corrected this to this because this.
试着让LM中的所有过程更加可见。让它报告。我把这个改成这个是因为。

01:26:04
Right? I saw these words in the context. And so therefore, it sounds like this errors could be that.
对吧?我在文中看到了这些词。因此，听起来错误可能是这样的。

01:26:10
So using it in a way that makes its thinking slightly more legible.
所以用一种让它的思想更清晰的方式来使用它。

01:26:17
Um.
嗯。

01:26:26
Next colleague.
未来的同事。

01:26:26
Where they were testing, can tell.
他们在哪里测试，我能看出来。

01:26:30

If I'm going to ask you this question, I give you the statement, and you're going to tell me.
如果我要问你这个问题，我给你这个表述，你要告诉我。

01:26:37
Yes or no, right? So, yes, no question that food is born before the Roman and cohesion.
是或不是，对吧？所以，是的，毫无疑问，食物是先于罗马而诞生的。

01:26:45
Yes or no, right? We know. We know this right moment.
是或不是，对吧？我们知道。我们知道这是正确的时刻。

01:26:54
Emperor, clearly for the French.
皇帝，显然是法国人。

01:26:59
Right?
对吧?

01:27:00
E.G.?
例如呢?

01:27:00
Information that is very obvious to us. The airline has a sense of information because it's on things that the key data right? Both dates are there.
这些信息对我们来说是显而易见的。航空公司有一种信息意识因为它是关于关键数据的，对吧？两个日期都在这里。

01:27:11
But I am, gets it wrong, because the context around Napoleon is so distinct with context.

但是我错了，因为关于拿破仑的背景和背景是如此的不同。

01:27:18
You're not going.
你不能去。

01:27:21
Into the source.
进入源头。

01:27:22
We access technology for.
我们获取技术是为了。

01:27:27
And we made a context. It's because the context will roam so far from the context, from the front.
我们做了一个背景。这是因为上下文会漫游到离上下文很远的地方，从前面。

01:27:33
You cannot always make that link.
你不能总是把它们联系起来。

01:27:38
Something else like Angela Merkel is an alumnus of.
还有像安吉拉·默克尔这样的校友。

01:27:46
Let's see you track University, right?
让我们看看你追踪大学，好吗?

01:27:51
General Medical Aluminance of Okay.
一般医学知识。

01:27:54
Flip it around. Utrek University is where Indelam Local is a limits off.
把它翻过来。Utrek大学是Indelam Local的禁区。

01:28:02
The LM gets it right for the first version, which is a common version, it gets.
LM在第一个版本中得到了正确的结果，这是一个常见的版本。

01:28:08
It's wrong for the second version. Maybe you flip things around.
第二个版本是错误的。也许你把东西倒过来。

01:28:11
We know, because we're humans. We read it. We know they mean the same, which is just a matter of.
我们知道，因为我们是人类。我们读了。我们知道它们的意思是一样的，只是。

01:28:17
Switching the terms around, reconstructing the sentence, but because it's unusual, formally speaking.
转换术语，重新构建句子，但因为这是不寻常的，正式地说。

01:28:25
Okay, ellen, doesn't catch it.
好吧，艾伦，我没听懂。

01:28:28
You see, this is reason. Reason doesn't.
你看，这就是道理。没有理由。

01:28:32
Happen when L and it's all probably ballistic.
发生在L的时候，可能都是弹道。

01:28:35
What we want to share. So obviously I tried that and the GN AI, so I' used the open source one, so I was trying to pick up those ocr.
我们想要分享的。显然我尝试了那个和GN AI，所以我用了开源的那个，所以我试着拿起那些ocr。

01:28:49
Then you say, so this is a sentence I gave it as tests. Her parents were party pandas. So they actually went to highlight party, say that it's ag pandas. Panda pandas.
然后你说，这是我作为测试给它的一个句子。她的父母是派对熊猫。所以他们实际上去了一个突出的派对，说那是一只熊猫。熊猫熊猫。

01:29:01
Oh, interesting, because I ask for a reason, we can say the token party is like E and OC arrow for pumpers, party is a term used in some languages for rice.
哦，有趣的是，因为我问一个原因，我们可以说，代币党就像E和OC箭头对泵，党是一个术语在某些语言中用于大米。

01:29:10
But in the context of family background, planters is more causible. So planters come this.
但在家庭背景的背景下，种植园主更有可能。所以种植者来到这里。

01:29:16
Like a surname, is it?
就像姓一样，是吗?

01:29:19
Is because, like, it's a bit, it's a bit of bemmy. Background. Okay, okay. Okay. So you feel that's not right, right? Right? Rice paddy will be so far away from the.
因为，就像，有点，有点脏。背景。好吧,好吧。好吧。所以你觉得这不对，

对吧？对吧?稻田会离我们那么远。

01:29:31
From the context flow, that is family.
从上下文流来看，这就是家庭。

01:29:35
So these errors do pop up, right, and we.
这些错误会弹出，对吧。

01:29:43
This part is when you ask an LM to estimate how confident they are, they are always overcome.
这部分是当你要求LM估计他们有多自信时，他们总是被克服。

01:29:48
It will tell you real 90 %, 95 %.
它能告诉你90%到95%的真实情况。

01:29:54
Okay.
好吧。

01:29:54
They've done the same thing with humans. And it's interesting that humans, even if they're not as they are, what they are about, as equally accurate, but they are much more aware about what they don't.
他们对人类也做了同样的事情。有趣的是，人类，即使他们不是他们的样子，他们是什么，同样准确，但他们更清楚他们没有什么。

01:30:07
The humans have a better estimation of, okay, I might be wrong. I'm only got 60 %.
人类有更好的估计，好吧，我可能错了。我只有60%

01:30:13
And confident, whereas LM will be, I'm 90 % confident that I got.
自信，而LM则是，我有90%的信心。

01:30:18
That's right. They don't have a good estimation of So sometimes when you ask, LM How should I know about this? If I can.
这是正确的。他们没有很好的估计所以有时候当你问，LM我怎么知道这个?如果可以的话。

01:30:27
Um, not very.
嗯，不太好。

01:30:31
So you have to be quite careful.
所以你必须非常小心。

01:30:33
Um.
嗯。

01:30:34
Personally, I would just use element as an exploratory tool.
就我个人而言，我只是将element作为一种探索工具。

01:30:38
Because I think you can always kind of double check what they're saying a lot more than you can.
因为我认为你总是可以反复检查他们说的话。

01:30:46
Of a lot of other operations, my colleagues, who are we more into it?
在很多其他的行动中，我的同事们，我们对谁更感兴趣?

01:30:52
I can see why, in a sense. But there are also many things that we can do with other methods.
从某种意义上说，我明白为什么。但是我们也可以用其他方法做很多事情。

01:31:02
Like traditional text analysis tools that will achieve the same result of very similar results, while also being very much more replicable.
就像传统的文本分析工具一样，它们会得到非常相似的结果，同时也更容易复制。

01:31:10
And we can peek under the black box and see exactly why the answer is to, which is.
我们可以在黑盒子下面看看为什么答案是，也就是。

01:31:15
And it also doesn't have to data center cost and environmental costs and social issues.
它也不需要数据中心成本，环境成本和社会问题。

01:31:20
Right? So.
对吧?所以。

01:31:26
So in terms of working with HDR, ECR, right? It does take a lot of time and effort.
所以在与HDR和ECR合作方面？这确实需要花费大量的时间和精力。

01:31:39
So sometimes.
所以有时。

01:31:41

But with Jenny, I it might actually be easier to do it, not just the experience of the call.
但和珍妮在一起，我可能会更容易做到，不仅仅是打电话的经历。

01:31:51
The time cost.
时间成本。

01:31:52
So something to become more awesome.
所以要变得更棒。

01:32:02
Advance on your project. If you have a meeting, then doesn't.
推进你的项目。如果你有会议，那就不用了。

01:32:14
Youth machine automation for something, when Gindi sits in there is kind of your own decision, right?
年轻的机器自动化的东西，当金迪坐在那里是你自己的决定，对吧?

01:32:21
And it's interesting, I know, because I was working on.
我知道这很有趣，因为我一直在研究。

01:32:25
Perspective, problem.
的角度来看,问题。

01:32:27
Using an AI to potentially help with some of the corrections. Hopefully, you should see what you can.
使用人工智能来帮助纠正一些错误。希望你能看看你能做什么。

01:32:35

I mean, the Party one is already very, it's very illuminating because it tells you how Eurocentric.
我的意思是，第一个党已经非常，非常有启发性，因为它告诉你欧洲中心主义是如何的。

01:32:43
The caucus is the LM stream, but where do you can unit?
党团会议是LM流，但你可以在哪里单位?

01:32:50
In a way. But again, these are things that you don't expect. So if you don't expect them to not be able to notice, then how do you foreground?
在某种程度上。但是，这些都是你意想不到的。所以如果你不希望他们注意不到，那么你怎么做前景呢?

01:33:00
How do you anticipate that your time?
你预计你的时间会怎样?

01:33:04
All right, um.
好吧，嗯。

01:33:09
We talked about this.
我们讨论过这个。

01:33:12
This neural network applies the same for text and images, right? It's the same concept.
这个神经网络同样适用于文本和图像，对吧？概念是一样的。

01:33:20
But neural networks are not necessarily genii. This is where the line kind of blows a little bit. An image classification algorithm.

但是神经网络并不一定是天才。这条线有点乱了。一种图像分类算法。

01:33:30
Okay, let me rephrase that all LMS are of algorithms.
让我重新表述一下所有LMS都是算法。

01:33:37
It's just out, and.
它刚刚出来，而且。

01:33:40
Predict what comes next. Algorithm can do several things. Algorithms can give you the next best predicted choice. You can also do classification, like.
预测接下来会发生什么。算法可以做很多事情。算法可以给你下一个最佳的预测选择。你也可以做分类，比如。

01:33:50
Yes. So they're called generated AI because they generate things for you, right? Whereas algorithms like this.
是。所以它们被称为生成人工智能因为它们为你生成东西，对吧？而像这样的算法。

01:33:59
On the under the hood, kind of share breaks in the architecture, but they're meant to just help you do tasks like possible.
在底层，在架构中有一些共享，但它们只是为了帮助你完成可能的任务。

01:34:05
Is that a cat?
那是一只猫吗?

01:34:06
It's an animal.
它是一种动物。

01:34:09
You don't generate the same way as gender.
你和性别产生的方式不一样。

01:34:13
But underlying structure, the same the neural networks and weights.
但是底层结构，神经网络和权重是一样的。

01:34:19
They tell you, okay, week two, triangles, more than zero triangles.
他们告诉你，好的，第二周，三角形，超过0个三角形。

01:34:25
It should be at least one time.
至少应该有一次。

01:34:26
And what's interesting is, if you look at the research that's going on in a moment when an LM, or in this case an algorithm, has been trained, it can be adapted part of the.
有趣的是，如果你看一下正在进行的研究当一个LM，或者在这个例子中是一个算法，被训练后，它可以适应一部分。

01:34:41
So I think I mentioned Matt Reader before. Matt Reader is a project that I was falling very closely on.
我想我之前提到过马特·里德。《马特·瑞德》是我一直密切关注的一个项目。

01:34:50
This is the one that looked at all the maps of the UK, from their National Survey to identify railway lines every.
这是一个研究了英国所有地图的人，从他们的国家调查中找出了每条铁路线。

01:35:00
Through a weight infrastructure automatically, right? So they train it a little bit. They took an off the shelf Google algorithm, right?
自动通过权重基础设施，对吧？所以他们稍微训练了一下。他们采用了现成的谷歌算法，对吧?

01:35:09
And then they said, okay, now look for this. They got a concept from a tumor detection software.
然后他们说，好吧，现在看看这个。他们从一个肿瘤检测软件中得到了一个概念。

01:35:18
Right? There's tumor detections available for normally simple specifically.
对吧?有一些简单的肿瘤检测方法。

01:35:23
So they kind of do inspiration, combine them, retrain an algorithm to look for specific things, like railway lines, right? The hedge lines.
所以他们做一些灵感，把它们结合起来，重新训练一个算法来寻找特定的东西，比如铁路线，对吧？树篱的线条。

01:35:33
Or specific diagrams. So we've gone the same way. It takes way less effort and time and energy to.
或者具体的图表。所以我们走了同样的路。这需要更少的努力、时间和精力。

01:35:42
We train something, because actually, the wits here are the same. Their infrastructure is the same. The wits are changed. You just shift the wits a little bit.
我们训练一些东西，因为实际上，这里的智慧是一样的。他们的基础设施是一样的。智慧变了。你只是稍微转移一下思维。

01:35:52
Hope. Look for something, rather than a cat.
希望。找点东西，而不是找只猫。

01:35:55
This respect to this idea of minimal computing, right adapting rather than creating construction.
这是关于最小计算的想法，是适应而不是创造结构。

01:36:06
And I know some of you are working on tools around image recognition.
我知道你们中的一些人正在研究图像识别工具。

01:36:17
And training. And actually, a lot of the cognitive book, right that you do as historians is.
和培训。事实上，很多关于认知的书，对吧，作为历史学家，你们所做的是。

01:36:23
Doing the design, doing the project, doing the annotated data set right, rather than.
做好设计，做好项目，做好标注数据集，而不是。

01:36:31
Working the algorithm itself, necessarily.
必须自己操作算法。

01:36:36
And so that's why, for at least for this course, I'm happy for you to go.
这就是为什么，至少对于这门课，我很高兴你们能去。

01:36:42
Energy that the inside, there's Q that is aligned to what you will eventually

do with. And tell anyone.
内部的能量，这里是Q与你最终要做的是一致的。告诉别人。

01:36:49
Oh.
哦。

01:36:50
You just do exploring and thinking about how you want to scope it.
你只需要探索和思考你想要如何确定它的范围。

01:36:55
All right. So we've been looking at corpuses, all right sources and corpuses that we need to.
好吧。所以我们一直在看语料库，好的，我们需要的来源和语料库。

01:37:10
Explore, interrogate, there's some ph tools that refuse AI tools that are kind of working, E.G.?
探索，审问，有一些ph工具拒绝使用人工智能工具，例如?

01:37:17
Right? So Deligan spots, E.G., it's about quering the material, right?
对吧?所以德里根指出，例如，这是关于对材料的质疑，对吧?

01:37:22
But we want to do a slight segway into quantitative history. There's a reason why I put it together, and it's because you share very fundamental assumptions about how.
但我们想稍微深入一下定量历史。我把它们放在一起是有原因的，因为你们对如何。

01:37:33
We can use evidence of the world.
我们可以使用世界的证据。

01:37:36
And you should pass and solving.
你们应该通过并解决。

01:37:40
You've really seen some of this quantification when it came to say, text analysis, right? What count? Frequency counts? Those are quantitative.
当谈到文本分析时，你们确实看到了一些量化，对吧？计算什么?频率计数?这些都是定量的。

01:37:50
You're counting the number of instances to work, say democracy happens from 1 billion models, or the number of times to where children appears.
你在计算起作用的实例的数量，比如民主从10亿个模型中发生，或者儿童出现的次数。

01:37:59
And what sticks to it.
还有粘在上面的东西。

01:38:01
So it's not a new thing, but as a field, it really takes off in the sixties to grab your.
所以这并不是一个新事物，但是作为一个领域，它在六十年代开始发展起来，抓住了你的。

01:38:09
So at the Anderson Reading, right, which is a historic rock historical review about quantity.
在Anderson Reading，这是一个关于数量的历史回顾。

01:38:16
Different pieces of view, right? And this is, there was a very, very big kind of optimism that we can count away.

不同的观点，对吧？这是，有一种非常，非常大的乐观，我们可以忽略不计。

01:38:26
To good history.
敬美好的历史。

01:38:29
We know enough data about people in the past, we will be able to identify, kind of, where the sources are silent. So history from below, right? If we know how much something costs, then we can recreate.
我们对过去的人有足够的了解，我们将能够识别，在某种程度上，信息源是沉默的。下面的历史，对吧？如果我们知道某样东西的成本，那么我们就可以重新创造。

01:38:42
The life of a person.
一个人的一生。

01:38:45
Or factually broken if there are no direct sources of the problem. All right.
或者，如果没有直接的问题来源，实际上已经破裂了。好吧。

01:38:50
Totalizing aspect to quantitative history, right where we can create and aggregate And maybe we'll shed some light.
量化历史的综合方面，就是我们可以创造和汇总的地方也许我们会有所启发。

01:38:59
Help us extend the reach of our sources, right? So they used to call it geometrics.
帮我们扩大线人的范围，对吧？所以他们以前叫它几何。

01:39:06

But of course, the concept here is you're moving from a single event to sort of a larger education, right?
但当然，这里的概念是你正在从一个单一的事件转移到一个更大的教育，对吧?

01:39:14
And the example that Anderson gives is a presidential election.
安德森举的例子是总统选举。

01:39:20
I know a lot of us look at one event, and then we look at the context behind it. We look at the effects, the impact of the antecedent factors, right? What lets the result.
我知道我们很多人看一件事，然后看它背后的背景。我们看影响，看前因的影响，对吧？什么让结果。

01:39:31
But of course, when they did a history, we'll just use that as one data point, and then look at all other presidential elections and see what's actually comparable between them.
当然，当他们做历史时，我们只会把它作为一个数据点，然后看看所有其他的总统选举，看看它们之间有什么可比较的。

01:39:40
There's a very social science approach you can already tell, and I know some of us will recoil, because how can you compare, say, the 2016 election in the US to the 1912 one.
这是一种非常社会科学的方法，你已经可以看出，我知道我们中的一些人会退缩，因为你怎么能把2016年的美国大选和1912年的选举进行比较呢?

01:39:52
Right? Completely different contexts, different people, different times, different debates. Why should one have a relationship with the other other, other than the fact that they are?
对吧?完全不同的背景，不同的人，不同的时间，不同的辩论。为什么一个人

要和另一个人有关系呢，除了他们是彼此的事实?

01:40:04
Both presidential elections. And so there's a physiological can go to it that, I think it's why quantitative the treat never read.
两次总统选举。所以有一个生理上的原因，我认为这就是为什么定量治疗从来没有读过。

01:40:14
Picks up pixel, that's a few, but it is embedded.
拾取像素，这是少数，但它是嵌入的。

01:40:19
And then we, has gotten the Second Life because of.
然后我们得到了第二人生，因为。

01:40:23
With this, it's your GS dsset. Anyone. JAS dset. You've got the rules. Your thesis, right location or specific.
有了这个，它就是你的GS dsset。任何人。雅dset。你有规则。你的论文，正确的位置或具体。

01:40:37
Thing, a polygonal line, and then you've got your attributes, right? So this is, let's say, your ID, right?
多边形线，然后是属性，对吧？这是你的ID，对吧?

01:40:45
The name of the thing, and then you've all these different attributes.
东西的名字，然后你有所有这些不同的属性。

01:40:49
So we've got, let's say, the Netherlands, it is a country. It has a polygon, it has a partner skill number, so on and so forth. Right? These are all the attributes.

比如说，荷兰，这是一个国家。它有一个多边形，它有一个伙伴技能数，等等。对吧?这些是所有的属性。

01:41:01
Right? The additional information that's related to your case, right? Or your specific unit.
对吧?和你的案子有关的额外信息，对吧？或者你的特定单位。

01:41:08
Analysis for industries countries. So a country has attributes like land error, population data. Founding.
工业国家分析。一个国家有土地误差，人口数据等属性。成立。

01:41:17
Blah, blah, blah blah, right? And you use the name or the keys to organize all the information.
废话，废话，废话，对吧？你用名字或键来组织所有的信息。

01:41:25
Right? So you can say, compare when countries performed around the world.
对吧?所以你可以说，比较世界各国的表现。

01:41:31
Hundred and 90 countries in the border from Beijing. You can do that.
与北京接壤的一百九十个国家。你可以这么做。

01:41:36
When were most countries founded, right?
大多数国家是什么时候建立的?

01:41:40
Hope.
希望。

01:41:45
Your unit analysis.
你的单位分析。

01:41:51
Maybe you've got the correct attribute, like population sites.
也许你已经得到了正确的属性，比如人口聚集地。

01:41:55
And this is where, if you look at quantitative history, the importance of documentation reappears, and this is something that we.
这就是，如果你看一下定量的历史，文件的重要性再次出现，这是我们。

01:42:12
Have been talking about, right, documenting your choices when it comes to your data. So if, let's say you are recording the length of.
我们一直在讨论，当涉及到数据时，记录你的选择。所以如果，假设你记录的是。

01:42:22
A country's borders. Do you use kilometers? Do you use meters? Do you run off? Or do you run off?
一个国家的边界。你用公里吗？你们用计价器吗？你会逃跑吗？还是你会逃跑?

01:42:28
Right, if you're counting, um.
好吧，如果你在数的话。

01:42:32
Split on language.
语言上的分歧。

01:42:37
Less in Singapore. Okay.

新加坡的情况更少。好吧。

01:42:41
Singapore. What do you think is our dominant first English?
新加坡。你认为我们最主要的第一种英语是什么?

01:42:45
I heard English. It's not.
我听到了英语。它不是。

01:42:51
Because I teach Mandarin, because our mother tongue is recorded. It's our first language.
因为我教普通话，因为我们的母语被记录下来了。这是我们的第一语言。

01:42:57
English is not considered the mother tongue of most people in Singapore.
英语不是新加坡大多数人的母语。

01:43:02
This is a policy decision, and it's very interesting. This is why, if you're in Singapore, you're in Singapore, or you're studying in the Singapore Institution.
这是一项政策决定，非常有趣。这就是为什么，如果你在新加坡，你在新加坡，或者你在新加坡学院学习。

01:43:13
You go to see the UK or the US, you still need to do a language test, because Singapore is not a majority English speaking country.
如果你去英国或美国，你仍然需要做语言测试，因为新加坡不是一个以英语为主要语言的国家。

01:43:20
Statistically, our mother tongue is our first language, not English, right? So you have the same issue, which is very funny, because you look at the

lesson like.
从统计数据来看，母语是我们的第一语言，而不是英语，对吧？所以你有同样的问题，这很有趣，因为你把这堂课看成是。

01:43:31
Jamaica is an English speaking country. Let's think why is not right?
牙买加是一个讲英语的国家。让我们想想为什么是不对的?

01:43:36
So the way that the statistics are conflict, the way that people are classified, is very interesting, and it does have.
所以统计数据的冲突，人们被分类的方式，是非常有趣的，它确实有。

01:43:46
Impact in terms of the way you should evaluate the data. Because if you look at just on the surface level, right, like, what is the mix now? 70 % Chinese, so 70 %.
影响你评估数据的方式。因为如果你只看表面，对吧，就像，现在的混合物是什么？70%是中国人，所以是70%。

01:43:57
Mandarin, nice as possible. Mother Tongue, right? 20 %. Molly, 10 %.
普通话，越好越好。母语，对吧？20%。莫莉，10%。

01:44:02
Well, indian sub population is different. It's mostly camel but convenient many other languages.
嗯，印度的亚人口是不同的。它主要是骆驼，但方便许多其他语言。

01:44:10
Then that changes the configuration. It changes your understanding of if you're just looking at those numbers, whether you think this country speaks English.
这就改变了构型。它改变了你的理解如果你只看这些数字，你是否认为这个国家说英语。

01:44:18
At all. And if each country has a different decision about this, this is a reliable dataset for you to understand the use of, say, english, accountable.
在所有。如果每个国家对此有不同的决定，这是一个可靠的数据集，让你了解使用，比如说，英语，问责制。

01:44:28
But we only know this because we know the context.
但我们知道这些是因为我们知道背景。

01:44:34
Against It is rare that his foreign comes back in. You know, enough about the complex.
他的外国人很少回来。你知道，说够了这个情结。

01:44:44
Judge whether this data set is good enough or whether you can depend on it if you are just going to run some calculations off the numbers.
判断这个数据集是否足够好，或者你是否可以依赖它，如果你只是要对数字进行一些计算。

01:44:52
The calculations would be correct, will be done correctly, but your original data is not representative.
计算将是正确的，将正确地完成，但您的原始数据不具有代表性。

01:44:58
Then you're going to have to.
那你就得。

01:45:00
And so then that's your biggest challenge, right? Knowing enough about this idea of a warrior in this keeping.
这就是你最大的挑战，对吧？对这个堡垒里的战士有足够的了解。

01:45:11
To know, actually whether the data you're looking at is truly representative.
要知道，你所看到的数据是否真正具有代表性。

01:45:16
What is the level of uncertainty that you have with the data?
你对数据的不确定程度是多少?

01:45:26
And what is?
什么是?

01:45:28
What you're doing.
你在做什么。

01:45:31
That's a tough question, and it takes a lot of especially when you're working at skill, you need to understand the context so well that you can using the.
这是一个很难的问题，它需要很多尤其是当你在练习技巧的时候，你需要很好地理解上下文，这样你才能使用。

01:45:45
The dataset, like you can this for.
数据集，就像你可以这样。

01:45:47
But once you do that.
但是一旦你这么做了。

01:45:52
And.
和。

01:45:53
So it was good enough of and enough for you.
所以这对你来说已经足够好了。

01:45:56
Your purposes, it might not be good enough for some other purposes, but if, for you, it's okay, then you can actually do a lot of interesting analysis.
你的目的，它可能对其他目的不够好，但如果，对你来说，它是可以的，那么你实际上可以做很多有趣的分析。

01:46:02
But, of course, we want to be very honest to say we are happy with this level of uncertainty.
但是，当然，我们要非常诚实地说，我们对这种程度的不确定性感到满意。

01:46:13
Right? Whether it's kilometers or whether it's run off to the nearest kilometer, doesn't really matter for us.
对吧?不管是千米还是最近的千米，对我们来说都不重要。

01:46:20
Exactly the line doesn't really, it doesn't, really isn't going to change our conclusion.
这条直线不会，不会，不会改变我们的结论。

01:46:25
Doesn't impact our research question, right?
不会影响我们的研究问题，对吧?

01:46:29
I just put a little line here about the ecological fallacy, and listen brings up. And here they're talking about how it can be problematic to infer an individual's behavior.
我在这里写了一小段关于生态谬论的内容，听一下。在这里，他们谈论的是

如何推断一个人的行为是有问题的。

01:46:45
Based on just an education, a nationwide or community, wide education.
基于仅仅一种教育，一种全国性的或社区的、广泛的教育。

01:46:49
So we kind of do this a little bit in our usual world. We're like, we don't know what this person would think, but someone else in the rough sing region in.
我们在日常生活中也会这样做。我们就像，我们不知道这个人会怎么想，但是在粗糙的区域的其他人。

01:47:00
We're probably the same age. Our reading the same things. We have thought this way. We have this evidence, so analogically to be suspect, this person might have thought the same.
我们可能是同龄人。我们的阅读是一样的。我们是这样想的。我们有这个证据，所以类比地说，这个人可能也有同样的想法。

01:47:09
Right, but that's because we are working with tough evidence that.
没错，但那是因为我们有确凿的证据证明。

01:47:13
It's not comprehensive enough. But.
它不够全面。但是。

01:47:16
You can't.
你不能。

01:47:22
Um.
嗯。

01:47:35
So with the data set, and we're thinking about, so British representatives or not, right? Sometimes it's really about thinking about the sheet.
有了这个数据集，我们在考虑，英国代表与否，对吧？有时候真的是想着床单。

01:47:49
What is a couple?
什么是情侣?

01:47:50
What are some of the parameters that has that we can tell you?
我们能告诉你的一些参数是什么?

01:47:58
All the society that you're looking at. It's a demographic one. It's a lot of population. It's about literacy.
你看到的整个社会。这是一个人口统计学问题。它的人口很多。这是关于读写能力。

01:48:07
Is about literacy. Can it tell you something about technology if you are interested in technology?
是关于读写能力。如果你对技术感兴趣，它能告诉你一些关于技术的东西吗?

01:48:14
The limit of the asset? What's a sheet of it?
资产的限额？一张纸是什么?

01:48:19
Seen multiple things.
见过很多东西。

01:48:22
And actually that is where the human comes in.
实际上这就是人类的作用所在。

01:48:33
Okay.
好吧。

01:48:33
On the.
在。

01:48:36
I put in this part here because I remember this vividly from where we did GS. Remember where we did GS I give you this map of the hub.
我把这部分写在这里是因为我清楚地记得我们做GS的时候。还记得我们做过的GS吗，我给你们这张枢纽的地图。

01:48:47
Right? And I said, okay, now plot points, right. And points, the GCP, the ground control points right, where you use.
对吧?我说，好的，现在是情节点。还有点，GCP，地面控制点，你用。

01:48:56
You see something on the map, and you point to the actual world, and you say these two other things. And then I heard a lot of confusion. How do I know which one do I pick? All right.
你在地图上看到一些东西，然后指着现实世界，然后说另外两件事。然后我听到了很多困惑。我怎么知道该选哪个呢？好吧。

01:49:05
And that's a constant question you kind of ask, you know, enough about the world to make this education, to say, this point on this map, is this point in the world Here.
这是一个你经常会问的问题，你知道，关于这个世界，足够多的东西来做这

个教育，说，这张地图上的这个点，是世界上的这个点。

01:49:15
Right? Same thing for seven, six, right.
对吧?7 6也是一样。

01:49:19
This number truly does represent. This reality is part of reality.
这个数字确实代表了。这个现实是现实的一部分。

01:49:25
When this is 70 % Chinese speaking in Singapore, is this reflective of English speaking ability in this country?
当新加坡70%的人说中文时，这是否反映了这个国家的英语口语能力?

01:49:34
Do you know enough to say that? Yes, that's correct or no. It would be more complicated than that. The number of.
你有资格这么说吗？是，正确与否。实际情况要比这复杂得多。的数目。

01:49:42
I need this decision and my fiction.
我需要这个决定和我的小说。

01:49:48
This all.
这一切。

01:49:50
My head shifted this building. My head moves. The left holds Man.
我的头移动了这座大楼。我的头在动。左边是人。

01:49:58
Thank you.
谢谢你！

01:49:58
Very sorry.
非常抱歉。

01:50:03
So to know enough for the world to understand whether that it doesn't make sense.
所以要知道的足够多，让世界明白它是否没有意义。

01:50:11
Everybody else land spots for you.
其他人都来找你。

01:50:18
Not easy.
不容易的。

01:50:18
But the good thing is you probably wouldn't even be less in the fields that you really know, right? You're interested in questions in areas of some history that you.
但好的一面是，你在你真正了解的领域可能不会少，对吧？你对某些历史领域的问题感兴趣。

01:50:33
Really somewhat familiar with and you know how to double check these things, right?
对这些很熟悉，也知道如何反复检查，对吧？

01:50:38
There were people party farmers in this period. Is it normal for people to be party farmers in this period?
这一时期出现了党农。在这个时期，人们做党农是正常的吗？

01:50:46
Does it make sense that it ought to be hardy and not planted?
它应该是耐寒的而不是种植的，这有意义吗?

01:50:50
Probably because you have the complex.
可能是因为你有复合体。

01:50:52
And so if anyone, you know, one knows about it, probably you would know. And even if you don't know where to find information.
所以如果有人知道这件事，你可能会知道。即使你不知道去哪里找信息。

01:51:02
We're just doing kind of that cognitive work and figuring out back checking, double checking, exploring that context.
我们只是在做一些认知工作，然后再检查，再检查，探索那个环境。

01:51:11
Okay. Can also about, yes, this qualiative history, something that economic historians are a really good way.
好吧。对，这种定性历史，是经济历史学家的一个很好的方法。

01:51:21
Like, I will expect them to also be able to, like, predict the data that you see. And, but there's an overlap with them. That's a very, very, very, very big overlap.
比如，我希望他们也能预测你看到的数据。但是它们之间有重叠。这是非常，非常，非常，非常大的重叠。

01:51:33
In economic history, and the fact that they did the truth. Well, if they did the truth, it was also slightly bigger. You're listening pizzas like population and homographic.
在经济史上，以及他们所做的事实。好吧，如果他们说的是实话，它也会稍

微大一些。你听到的披萨像人口和同源。

01:51:41
Okay, um, some GS, what is also powerful as well.
好的，一些GS，也很强大。

01:51:45
That makes sense. I learned GYS more publishing story.
这说得通。我了解了GYS更多的出版故事。

01:51:50
Because it's key. It's important to be their work. And.
因为这是关键。成为他们的工作很重要。和。

01:51:56
Showing demographic change over time, or access to minerals or particular locations.
显示人口随时间的变化，或矿产或特定地点的获取。

01:52:02
Right? And so you'll find that some disputes of.
对吧?所以你会发现一些争论。

01:52:07
But economic concerns being very happy working with evidence that could be good.
但经济方面的担忧是非常高兴的，因为证据可能是好的。

01:52:15
And there's a sense in recent economic history that there are some terms that may be more transferable than that.
在最近的经济史上有一种感觉，有一些术语可能比这更容易转移。

01:52:24
It's insight.

的洞察力。

01:52:25
Of love, treat volume. These are numbers that you can kind of pin down a bit more securely than things like conceptions about gender.
对爱，对待量。这些数字比性别概念之类的东西更可靠。

01:52:38
Okay.
好吧。

01:52:38
But it's going to be like, they're being said, like, I think, even for economic historians, like there are also, like, the critical element is still there because, like, there are a lot of shifts, even.
但就像，他们说的，就像，我认为，即使对经济历史学家来说，也有，关键因素仍然存在，因为，甚至有很多变化。

01:52:49
In the field where, like, there are more criticals of data that they have. So, like, for the longest time, like GDP was taken as, like, the measure of, like the wealth of a state, right? And then, in like, the last 1015 years, about a new model. But in other people, even they're going to use the kind of.
在这个领域，他们有更多的关键数据。所以，在很长一段时间里，GDP就像是，一个国家财富的衡量标准，对吧？然后，在过去的1015年里，出现了一个新的模型。但对其他人来说，即使他们也会使用这种。

01:53:02
Which are like, okay, all these data we're getting on, like, GDP of countries, is that actually representative of the population? Like poverty studies, as the big public studies is a big sum in economic history, because.
就像，好吧，我们得到的所有这些数据，比如，各国的GDP，这真的能代表人口吗？像贫困研究一样，作为大型公共研究在经济史上占有很大的份额，因为。

01:53:12
The site will be like, okay, let's look for other data sets of our constitutes, like the actual lived experience of an economic of people living in an economy. And then, like in ancient Romans, it is actually like there's also a recent shift where.
这个网站会说，好吧，让我们找找其他的数据集，比如一个经济体中人们的实际生活经验。然后，就像在古罗马一样，实际上最近也有一个转变。

01:53:22
I mean, you can talk to Documentomia about how, like, his whole thing, about how well. Now, you know, in the past, people used to like, assume the Roman state operated like modern economies, where the historians are all looking at life.
我是说，你可以跟文献展谈谈，他的整件事，他有多好。现在，你知道，在过去，人们习惯认为，罗马国家像现代经济一样运作，历史学家都在研究生活。

01:53:33
Official sources, or like thinking of the Roman economy operating as a very common control state, given thing. And then now archaeological evidence and new methods are kind of like reviewing what?
官方资料，或者把罗马经济看作是一个非常普遍的控制国家。现在考古证据和新方法有点像回顾什么?

01:53:43
Experts. We really new data sets, which people are using the conceptualize the normal economy differently. Now.
专家。我们有新的数据集，人们用不同的方式来定义正常经济。现在。

01:53:48
So it does import a lot of assumptions that we have about, even things are, how economies are run or understood.
所以它确实引入了很多假设，关于经济是如何运行或被理解的。

01:53:57

Euros. It's a bit.
欧元。有点。

01:53:59
Not for an echronistic. Maybe it's prone to an mechanism, because the comparison is key to the discipline.
不是为了时间。也许它倾向于一种机制，因为比较是这门学科的关键。

01:54:11
Making that comparison is key. So you kind of need to make a judgment in terms of whether it's a good comparative GDP is certain.
进行比较是关键。所以你需要判断它是否是一个好的比较GDP。

01:54:19
Measures that transferable between contexts. It could be that GD is really good for modern history, but very bad.
可在上下文之间转换的度量。也许GD对现代历史确实有好处，但非常糟糕。

01:54:28
Right?
对吧?

01:54:28
And that's your condition.
这就是你的状况。

01:54:29
Okay. Any questions.
好吧。任何问题。

01:54:35
And this is a bit heavy. You go through a lot today.
这个有点重。你今天经历了很多。

01:54:45

Okay, so this one we did here, we give some thought as to how AI can be a tool for our historical work.
好的，我们在这里做了一个，我们思考一下人工智能如何成为我们历史工作的工具。

01:54:57
And what happens is we train you as a method.
我们把你训练成一种方法。

01:55:00
Sure you'll learn somewhere in terms of how you feel about using GENI.
当然，您将在某处了解使用GENI的感觉。

01:55:06
Process. That means some good ways. That means something more.
的过程。这意味着一些好的方法。这意味着更多的东西。

01:55:15
Ultimately.
最终。

01:55:16
In your holiday or uncertainty? What exactly are you working with?
在你的假期或不确定？你到底在做什么？

01:55:25
That.
那。

01:55:25
Think about verification and responsibilities that you can remember of you with.
想想那些你能记住的证明和责任。

01:55:33

The story as a critical leader of sources and user of tools and methods, right?
这个故事是作为资源的关键领导者和工具和方法的使用者，对吧?

01:55:38
Evaluating these tools as a key skill.
评估这些工具是一项关键技能。

01:55:41
And then we can't put a longer view quickly and a larger view about a wider view of what data sets right that you've all been working with.
然后我们就不能从更长远的角度和更大的角度来看待你们一直在研究的数据集是正确的。

01:55:53
Point, right? And thinking about how they might also be considered thought of as devaluators, right? Where the rules are specific cases and the attributes are actually what you want to compare Things.
点,对吗?想想它们也可以被认为是贬值者，对吧？其中规则是特定的情况，属性实际上是你想比较的东西。

01:56:06
Right?
对吧?

01:56:07
They don't just give you a similar information. That information themselves are useful.
它们不只是给你类似的信息。这些信息本身是有用的。

01:56:13
Okay.
好吧。

01:56:13

And illustrate things beyond the sources themselves. That's the power of having a PR people, the PR matrix.
并说明资料本身之外的东西。这就是公关人员的力量，公关矩阵。

01:56:24
Set same thing as gis, E.G., the reason why we do GS is you can put different sources of information.
和地理信息系统一样，比如，我们做地理信息系统的原因是你可以放入不同的信息来源。

01:56:33
Geographical population, conceptual together, just for them in the same spot.
地理上的人口，概念上的在一起，只是为了他们在同一个地方。

01:56:40
And that really enhances our competition.
这确实提高了我们的竞争力。

01:56:42
Right? So a lot of these methods and goods are inherently comparative. The task for assets and sciences to consider whether that comparison is more difficult.
对吧?所以很多方法和商品本质上是比较的。资产和科学的任务是考虑这种比较是否更加困难。

01:56:52
And that usually boils down to context.
这通常归结为上下文。

01:56:55
And you're familiar.
你们很熟悉。

01:56:58

Any questions.
任何问题。

01:57:01
Okay.
好吧。

01:57:04
All right, cool. Next we're going to do visualizations, so we'll, I guess we'll continue a theme around tables and.
好吧，酷。接下来我们要做可视化，我想我们会继续围绕桌子的主题。

01:57:14
The way that we perceive information and output, right? Our visualization is like L end is a way of presenting, representing information.
我们感知信息和输出的方式，对吧？我们的可视化就像L端是一种呈现信息的方式。

01:57:23
And that means there are a lot of specific visual vocabulary to.
这意味着有很多特定的视觉词汇。

01:57:29
Starting to move to a Republican.
开始向共和党靠拢。

01:57:33
Angle might be a good.
角度可能很好。

01:57:35
In terms of how we should understand our outputs and how we.
就我们应该如何理解我们的产出以及我们如何。

01:57:42

They will be used and understood that position. It's still going to be data heavy as new behavior.
他们将被利用并理解这一立场。作为一种新的行为，它仍然需要大量的数据。

01:57:49
Because we will look at relations.
因为我们会研究关系。

01:57:51
And network analysis.
以及网络分析。

01:57:53
Right? Remembers of who's talking to who, who's in touch with who, right? We'll have a low look at what the kind of data looks like, and what kind of inputs come up to them.
对吧?记得谁在和谁说话，谁在和谁联系，对吧？我们会粗略地看一下这些数据是什么样的，以及它们需要什么样的输入。

01:58:06
All right.
好吧。

01:58:06
We got to go.
我们得走了。

01:58:08
Okay, oh.
好的,哦。